Inefficient Equilibria and Strategic Complementarities in Elections with Costly Information *

Carl Heese†

We revisit a standard model of costly information acquisition and information aggregation in large majority elections. A large electorate has common preferences, but voters are uncertain which of two policies is better and may acquire costly private information before voting (Martinelli, 2006). Two insights emerge. First, information acquisition exhibits strategic complementarities: when other voters acquire more information, a voter's own incentive to become informed may increase. We characterize when information acquisition is locally a strategic substitute and when it is locally a strategic complement. Second, the complementarities matter for efficiency: voters may coordinate on an efficient high-information equilibrium, but they may also coordinate on an inefficient low-information equilibrium. Rational ignorance of voters thus has a coordination component and is not purely driven by cost-benefit tradeoffs. We discuss implications for voter-information initiatives.

The Condorcet jury theorem and its modern versions (e.g. Austen-Smith and Banks, 1996; Feddersen and Pesendorfer, 1997) provide a central efficiency benchmark for democratic institutions: Even when relevant political information is dispersed among a large number of voters, majority elections lead to efficient outcomes under broad conditions. Piketty (1999) interprets this result as a political analog of the *First Welfare Theorem* for markets.

A challenge to the efficiency of political institutions is that political information is typically costly: In a large election, an individual voter has a negligible probability of affecting the outcome, and may therefore acquire too little information—or none at all—so efficiency can fail. This is the rational-ignorance problem emphasized by Downs (1957).

*. I am grateful for helpful discussions with César Martinelli and Faruk Gul.

†Corresponding author; The University of Hong Kong, HKU Business School, heese@hku.hk.

Martinelli (2006) made a fundamental observation about this problem, clarifying the conditions under which a version of the Condorcet jury theorem persists and "rational ignorance" does not preclude efficiency. Although each voter's private incentive to acquire information becomes small in a large electorate, the election may still aggregate many small pieces of information. His analysis formalizes the horse race between the growing number of voters and the shrinking amount of information acquired by each individual voter. If the marginal cost of arbitrarily imprecise information becomes arbitrarily close to zero, there exists an equilibrium sequence in which the election outcome is asymptotically efficient.

This paper revisits the setting of Martinelli (2006). In this setting, $2n+1$ voters have common state-dependent preferences over a binary policy. They prefer policy A in one state and policy B in the other. Before policy is decided with a simple majority vote, each voter can acquire a private signal with precision $x \geq 0$ at private cost $C(x)$.

The common interest suggests that information acquisition is a strategic substitute: the more information other voters acquire, the less incentive a given voter may have to become informed.

Our first result shows this intuition is incomplete. The acquisition of political information can *locally* exhibit strategic complementarities: a local increase in the information acquired by other voters can raise a voter's own incentive to become informed. In other words, at a given strategy profile, a voter's payoff is locally *supermodular* in her own information acquisition and the information acquired by others (cf. Topkis, 1979; Vives, 1990; Milgrom and Roberts, 1990). We characterize at which strategy profiles information acquisition is locally a strategic substitute and when it is locally a strategic complement (Proposition 1).

The strategic complementarities have important efficiency consequences: they drive the coexistence of multiple equilibrium sequences with different asymptotic properties. Our second result shows that the efficient equilibrium sequences identified by Martinelli (2006) coexist with inefficient equilibrium sequences in which the wrong alternative is elected with a probability bounded away from zero, even as $n \to \infty$ (Theorem 1). The costly-information version of the Condorcet jury theorem should therefore be read as a possibility result: efficiency obtains only when voters coordinate on the efficient equilibrium.

This coexistence is generic. Whenever an efficient equilibrium sequence exists, an inefficient equilibrium sequence exists as well, except in the knife-edge case in which voters are exactly indifferent between the two alternatives under the prior.

This knife-edge case is the main specification in Martinelli (2006). There, equilibrium incentives are governed purely by a strategic-substitute logic; as a result, the equilibrium with information acquisition is unique, and coordination is not an issue.[1]

The two types of equilibrium sequences differ sharply in their information aggregation properties. Along the efficient sequence, information aggregates: the realized vote share becomes informative about the state. Along the inefficient sequence, by contrast, information aggregation fails. An outside observer cannot consistently infer the state from the realized vote share as $n \to \infty$. This failure is not driven by high information costs. Perhaps surprisingly, inefficient low-information equilibrium sequences exist even when information is extremely cheap near zero; for example, when $C(x) = \exp(-1/x)$ for $x > 0$ and $C(0) = 0$ (Proposition 2).

The coexistence of low- and high-information equilibria is reminiscent of the low- and high-turnout equilibria in costly-participation models, such as Palfrey and Rosenthal (1983). As is well known, high-turnout equilibria in those models can be fragile to strategic uncertainty about preferences, costs, or the number of voters (Palfrey and Rosenthal, 1985; Myerson, 1998). In contrast, such refinements do not separate the two equilibria in our setting. In an extended version of our model with strategic uncertainty about preferences, information costs, and prior beliefs, the same coordination problem persists, except in a knife-edge case; see the companion paper (Heese, 2022).[2] This suggests that the multiplicity generated by costly information is qualitatively different from the familiar multiplicity generated by costly participation.

Related Literature. This paper contributes to the literature on voting with costly information acquisition. Prior work has emphasized the strategic-substitute aspect of political information: voters can free-ride on the information acquired by others. We show that information acquisition can also exhibit strategic complementarities and that this creates an equilibrium-coordination problem with sharp efficiency implications. In other words, rational ignorance in politics has a coordination aspect.

[1] Martinelli (2006) also considers asymmetric priors and preferences and shows that efficient equilibrium sequences continue to exist under the same conditions on the cost function as in the main specification.

[2] As usual in voting games, there is also an additional trivial equilibrium: all voters vote for the policy they prefer given the common prior, and no voter acquires any information. However, the trivial equilibrium is not robust: for example, in the extended model, it does not survive the requirement of undominated strategies, unlike the low-information equilibrium and the high-information equilibrium.

Most closely related is the literature on the Condorcet jury theorem with endogenous information acquisition; in particular, the line of work following (Martinelli, 2006). We have verified that our results extend to the model variations considered in the follow-up work, (see, e.g., Martinelli, 2007), Triossi (2013), and Oliveros (2013). Inefficient equilibrium sequences with information acquisition also exist in these variations. This implies necessary corrections to Theorem 3 in Martinelli (2007) and Proposition 3 in Triossi (2013). Other related Condorcet environments have been studied in Mukhopadhaya (2003); Koriyama and Szentes (2009) who show that larger-than-optimal committee sizes may lead to social-welfare losses relative to smaller committees because of free-riding, although these losses need not be large.

Other strands of the literature study related design questions (see, e.g., Persico, 2004; Gershkov and Szentes, 2009; Gerardi and Yariv, 2008). Because this literature typically focuses on efficient equilibria, our results motivate asking when additional nontrivial equilibria exist in these settings, possibly because of strategic complementarities in information acquisition. In such instances, questions of robust mechanism design naturally arise: For example, which voting mechanisms are worst-case optimal across equilibria?

An experimental literature has also evaluated testable implications of these theories; see, e.g., Grosser and Seebauer (2016); Elbittar *et al.* (2020); Mechtenberg and Tyran (2019); Bhattacharya *et al.* (2017). Our analysis invites follow-up work that tests the extent to which coordination problems matter empirically when voter information is endogenous.

A Cultural Interpretation. Following Schelling (1980) and Myerson (2009), one may interpret equilibrium coordination as a reduced-form representation of political culture. Societies may differ in the conventions, expectations, and focal points that make one equilibrium rather than another salient. In this sense, the low-information and high-information equilibria correspond to different cultures of political informedness.

Empirical work documents related differences in "cultures of news consumption." News consumption varies not only with individual preferences and information costs, but also with contextual norms (Toff and Kalogeropoulos, 2020). For example, some citizens follow political news because they regard keeping informed as part of civic duty (McCombs and Poindexter, 1983; Palmer and Toff, 2020). By contrast, other citizens exhibit a "news-finds-me" orientation: they expect important political information to reach them through social media, peers, or everyday exposure without active search (Gil de Zúñiga *et al.*, 2017, 2020). These findings suggest that political informedness is shaped partly by shared norms about whether citizens should actively follow politics, and that social media may be associated with more passive information norms.

This perspective also offers a possible interpretation of the limited effects of standard voter-information campaigns, as documented in the meta-analysis by Dunning *et al.* (2019). Our model suggests an explanation: if information provision affects individual information costs but leaves citizens' expectations about others' information acquisition and surrounding norms largely unchanged, the electorate may remain coordinated on the low-information equilibrium. Voter-information initiatives may therefore need to do more than provide facts or lower individual information costs: they may also need to foster a culture of political informedness, in which active engagement with political information is expected.

1 Model

We restate the model of Martinelli (2006), largely following its original notation. There are $2n+1 \geq 3$ voters, two policies A and B, and a binary state $z \in \{z_A, z_B\}$. The voters hold a common prior. The prior probability of state z_A is $q_A \in (0,1)$, and the prior probability of state z_B is $q_B = 1 - q_A$. Voters receive utility $U(d,z)$ if the outcome is d and the state is z. We denote $U(A, z_A) - U(B, z_A) = r_A$ and $U(B, z_B) - U(A, z_B) = r_B$ and assume that $r_A, r_B > 0$. Thus, all voters prefer A in state z_A and B in z_B. We assume $q_A r_A > q_B r_B$, so that policy A maximizes expected utility under the prior.[3]

The timing is as follows. Each voter chooses the precision $x \in [0, 1/2]$ of a binary private signal $s \in \{s_A, s_B\}$, where

$$\frac{1}{2} + x = \Pr(s_A \mid z_A) = \Pr(s_B \mid z_B).$$

The private signals are independent conditional on the state. Choosing precision x costs $C(x)$. The state and private signals then realize. After observing their private signals, all voters vote simultaneously, either for A or for B. Finally, the outcome is decided by simple majority rule.

[3] The case $q_A r_A < q_B r_B$ yields qualitatively the same results with the roles of A and B reversed. In the knife-edge case $q_A r_A = q_B r_B$, the low-information equilibrium sequence identified in the main result (Theorem 1) does not exist.

We assume that C is strictly increasing, strictly convex, twice continuously differentiable on $(0, \frac{1}{2})$ and that $C(0) = 0$; in other words, acquiring no information is costless. Note that $C'(0) \in [0, \infty)$ exists. Whenever $\lim_{x \downarrow 0} C''(x)$ exists, we write $C''(0) := \lim_{x \downarrow 0} C''(x)$, allowing $C''(0) = \infty$ when this limit diverges.

A pure strategy is a triple (x, v_A, v_B), where $x \in [0, 1/2]$ specifies the chosen precision, v_A specifies which policy the voter votes for after observing signal s_A, and v_B specifies which policy the voter votes for after observing signal s_B. A mixed strategy α is a probability distribution over pure strategies. We consider symmetric mixed-strategy Bayes–Nash equilibria. An equilibrium in which some strategy (x, v_A, v_B) with $x > 0$ is in the support is called an *equilibrium with information acquisition*. When no confusion arises, we simply write *equilibrium*.

2 Equilibrium Representation

We provide a compact characterization of equilibria with information acquisition in large electorates. To state the result, we first explain the "pivotal calculus" of strategic voters.

Fix a voter and the strategy α of all other voters. Whenever the votes of the other voters do not split into n votes for A and n votes for B, the fixed voter's choice of her strategy does not affect the outcome. Thus, when comparing expected utilities from two strategies, it suffices to consider the pivotal event in which the votes of the other voters split into n votes for A and n votes for B. Denote by $\pi(z; \alpha)$ the probability that a voter votes for A in state $z \in \{z_A, z_B\}$ under α. The probability of the pivotal event is

$$\Pr(\text{piv} \mid z; \alpha) = \binom{2n}{n} \Big(\pi(z; \alpha)(1 - \pi(z; \alpha)) \Big)^n. \tag{1}$$

Our result characterizes equilibria as strategies $\alpha(x, \delta)$ that mix between (x, A, B) with probability $1 - \delta$ and $(0, A, A)$ with probability $\delta > 0$ and satisfy two relevant equilibrium conditions.

Lemma 1 *Assume $q_A r_A > q_B r_B$ and $C'(0) = 0$. For all sufficiently large n, a strategy α is an equilibrium with information acquisition if and only if there is a* *pair* $(x, \delta) \in (0, \frac{1}{2}) \times (0, 1)$ *so that* $\alpha = \alpha(x, \delta)$ *and* α *satisfies*

$$\Big(\Pr(\text{piv} \mid z_A; \alpha) q_A r_A + \Pr(\text{piv} \mid z_B; \alpha) q_B r_B\Big) \left(\frac{1}{2} + x\right) - C(x) = \Pr(\text{piv} \mid z_A; \alpha) q_A r_A, \tag{2}$$

$$\Pr(\text{piv} \mid z_A; \alpha) q_A r_A + \Pr(\text{piv} \mid z_B; \alpha) q_B r_B = C'(x). \tag{3}$$

The formal proof builds on arguments already present in Martinelli (2006) and appears in the appendix. The first equilibrium condition states that voters are indifferent between (x, A, B) and $(0, A, A)$. To see this, compare the expected utilities from the two strategies. Up to the constant utility from non-pivotal events, the expected utility from choosing (x, A, B) is

$$\begin{aligned}\Pr(\text{piv} \mid z_A; \alpha) q_A \Big(U(A, z_A) \left(\frac{1}{2} + x\right) + U(B, z_A) \left(\frac{1}{2} - x\right)\Big) \\ + \Pr(\text{piv} \mid z_B; \alpha) q_B \Big(U(B, z_B) \left(\frac{1}{2} + x\right) + U(A, z_B) \left(\frac{1}{2} - x\right)\Big) - C(x).\end{aligned}$$

The expected utility from $(0, A, A)$ is, up to the same constant,

$$\Pr(\text{piv} \mid z_A; \alpha) q_A U(A, z_A) + \Pr(\text{piv} \mid z_B; \alpha) q_B U(A, z_B).$$

Subtracting $\Pr(\text{piv} \mid z_A; \alpha) q_A U(B, z_A) + \Pr(\text{piv} \mid z_B; \alpha) q_B U(A, z_B)$ from both expressions yields the stated indifference condition.

The second equilibrium condition states that the marginal cost of acquiring precision $x > 0$ equals the marginal benefit. Equivalently, it is the first-order condition for the optimal precision.

Under the mixed strategy $\alpha(x, \delta)$, a voter chooses (x, A, B) with probability $1-\delta$ and $(0, A, A)$ with probability δ. The induced voting probabilities are

$$\begin{aligned}\pi(z_A; \alpha(x, \delta)) &= (1 - \delta) \left(\frac{1}{2} + x\right) + \delta, \\ \pi(z_B; \alpha(x, \delta)) &= (1 - \delta) \left(\frac{1}{2} - x\right) + \delta.\end{aligned} \tag{4}$$

The prior bias toward A, captured by $q_A r_A > q_B r_B$, shapes the qualitative form of equilibria. It implies that all equilibria are shifted toward voting for A in the sense that all uninformed voters choose A. A prior bias toward B, i.e. $q_A r_A < q_B r_B$, would imply that, in all equilibria with information acquisition, the uninformed voters choose B. Thus, these equilibria involve mixing between (x, A, B) and $(0, B, B)$ instead of mixing between (x, A, B) and $(0, A, A)$.

In what follows, we identify equilibria with information acquisition by their representing pairs (x, δ).

3 Local Complements and Substitutes

The usual rational-ignorance argument emphasizes the strategic-substitute aspect of political information: voters can free-ride on information acquired by others. We now show that this logic is incomplete. Information acquisition can also be locally complementary.

We use a local notion because the relevant strategic effect changes sign across the strategy space. Fix the probability δ with which other voters remain uninformed and vote for A. Suppose that all other voters use $\alpha(x, \delta)$, and consider the payoff from choosing precision y and voting according to the signal. Up to terms independent of the voter's own action, this payoff is

$$\Psi_n(y; x, \delta) = \mathrm{MB}_n(x, \delta) \left(\frac{1}{2} + y\right) - C(y),$$

where

$$\mathrm{MB}_n(x, \delta) = \Pr(\text{piv} \mid z_A; \alpha(x, \delta)) q_A r_A + \Pr(\text{piv} \mid z_B; \alpha(x, \delta)) q_B r_B.$$

Thus, for $y' > y$,

$$\Psi_n(y'; x, \delta) - \Psi_n(y; x, \delta) = (y' - y)\mathrm{MB}_n(x, \delta) - \big(C(y') - C(y)\big).$$

The payoff gain from increasing own precision from y to y' is therefore increasing in the precision x used by other informed voters if and only if

$$\frac{\partial^2 \Psi_n(y; x, \delta)}{\partial y \, \partial x} = \frac{\partial \mathrm{MB}_n(x, \delta)}{\partial x} > 0.$$

We say that information acquisition is locally a *strategic complement (substitute)* at $\alpha(x, \delta)$ if

$$\frac{\partial \mathrm{MB}_n(x, \delta)}{\partial x} > 0, \text{ respectively, } < 0.$$

Hence our definition is the local analogue, for the voter's precision-choice problem, of the usual increasing-differences condition associated with strategic complementarities.[4]

The next result characterizes when information acquisition is locally a strategic complement and when it is locally a strategic substitute.

Proposition 1 *Suppose* $C'(0) = 0$ *and* $q_A r_A > q_B r_B$*, and consider the voters' strategy* $\alpha(x, \delta)$*. For every sufficiently large* n*, there exists a threshold* $\delta_{1,n} \in (0, 1/2)$ *such that*

1. *for* $0 \leq \delta < \delta_{1,n}$*, information acquisition is locally a strategic substitute at* $\alpha(x, \delta)$ *for all* $x \in (0, \frac{1}{2})$*;*
2. *for* $\delta_{1,n} < \delta < 1/2$*, information acquisition is locally a strategic substitute, then a strategic complement, and then again a strategic substitute, as* x *varies from* 0 *to* $\frac{1}{2}$*;*
3. *for* $1/2 < \delta < 1$*, information acquisition is locally a strategic substitute and then a strategic complement, as* x *varies from* 0 *to* $\frac{1}{2}$*.*

The proof is in the appendix. The marginal benefit of information is best understood through its connection to the *(expected) margin of victory* in the two states,

$$\begin{aligned} \mathrm{MV}(z; x, \delta) &= \left| \pi(z; \alpha(x, \delta)) - \frac{1}{2} \right| \\ &= \begin{cases} \delta + (1 - \delta) \left(\frac{1}{2} + x \right) - \frac{1}{2} & \text{for } z = z_A, \\ \left| \delta + (1 - \delta) \left(\frac{1}{2} - x \right) - \frac{1}{2} \right| & \text{for } z = z_B. \end{cases} \end{aligned}$$

The marginal benefit increases in the pivotal likelihoods $\Pr(\mathrm{piv} \mid z; \alpha(x, \delta))$, which are themselves monotone decreasing in the margin of victory, as

$$\Pr(\mathrm{piv} \mid z; \alpha(x, \delta)) = \binom{2n}{n} \left(\frac{1}{4} - \mathrm{MV}(z; x, \delta)^2 \right)^n. \tag{5}$$

Overall, a larger margin of victory implies a smaller marginal benefit of information.

We now explain the three regimes using this connection to the margins of victory.

[4]The terminology of strategic complements and substitutes is due to Bulow *et al.* (1985). The lattice-theoretic formulation of strategic complementarities in terms of increasing differences and supermodularity is developed by Topkis (1979), Vives (1990), and Milgrom and Roberts (1990); see also Milgrom and Shannon (1994) and Topkis (1998).

When $\delta = 0$, the margin of victory increases in x in both states. Thus, the value of information is monotonically decreasing, corresponding to substitutes. This monotonicity extends to values of δ in a whole interval $[0, \delta_{1,n})$, by continuity.

When $\delta_{1,n} < \delta < 1$, the vote share in z_A is larger than $\frac{1}{2}$ for all x but the vote share in z_B is larger than $\frac{1}{2}$ if and only if

$$x < x'' \text{ for } x'' = \frac{\delta}{2(1-\delta)}.$$

For $x < x''$, an increase in x has two countervailing effects: it increases the margin of victory in z_A, but it decreases it in z_B. The proof shows that the first effect dominates on an interval $[0, x']$, where $0 < x' < \min\{\frac{1}{2}, x''\}$, and the second effect dominates on $[x', \min\{\frac{1}{2}, x''\}]$. If $\delta > \frac{1}{2}$, then $x'' > \frac{1}{2}$ and these two intervals imply the overall substitutes-complements pattern. If $\delta < \frac{1}{2}$, there is another interval $[x'', \frac{1}{2}]$ on which the margin of victory increases with x in both states, implying an overall substitutes-complements-substitutes pattern.

Figure 1 below illustrates the three regimes in an example.

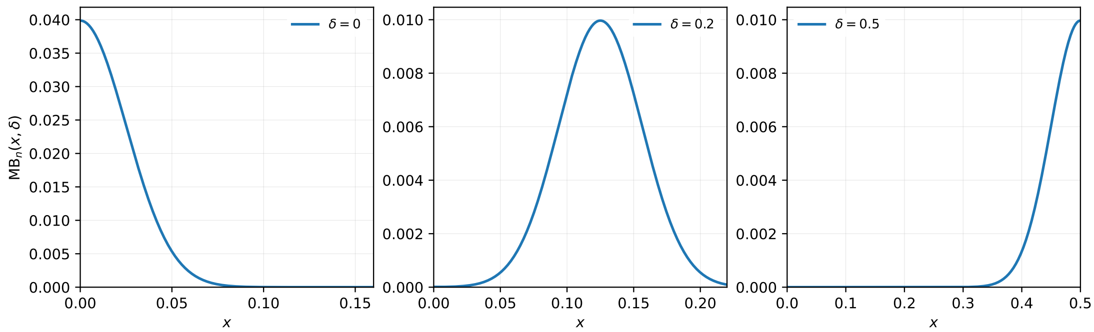

Figure 1: The marginal benefit $\mathrm{MB}_n(x, \delta)$ of information as a function of $x \in (0, \frac{1}{2})$ for fixed $\delta = 0$ (left), $\delta = 0.2$ (middle), and $\delta = 0.5$ (right), in an example with $n = 200$, $r_A = r_B = 1$, $q_A = 3/4$, and $q_B = 1/4$.

4 The Voters' Coordination Problem

We characterize the outcomes of large elections ($n \to \infty$). An equilibrium sequence $(\alpha_n)_{n \in \mathbb{N}}$ has *asymptotically efficient outcomes* if the probability that A is elected in state z_A and the probability that B is elected in z_B both converge to one as $n \to \infty$. It has *asymptotically inefficient outcomes* otherwise.

We focus on situations in which efficiency is possible. As shown by Martinelli (2006), a sequence of equilibria with asymptotically efficient outcomes exists if and only if $C'(0) = C''(0) = 0$. This condition means that the marginal cost of arbitrarily imprecise information is arbitrarily low. In the efficient sequence, individual information acquisition becomes arbitrarily imprecise, but slowly enough for the aggregate information—acquired by an increasing number of voters—to enable asymptotically efficient outcomes. Under the same condition, aggregate information costs converge to zero. Thus, the equilibrium sequence maximizes utilitarian welfare even after accounting for information costs (cf. Theorem 3 and Theorem 4 in Martinelli (2006)).

Our next result characterizes the asymptotic outcomes of all equilibrium sequences given $C'(0) = C''(0) = 0$. It shows that, whenever efficiency is possible, the voters face a coordination problem. They may miscoordinate on an equilibrium sequence with asymptotically inefficient outcomes.

Theorem 1 *Suppose $C'(0) = C''(0) = 0$ and $q_A r_A > q_B r_B$.*

1. *There exists a sequence of equilibria with asymptotically efficient outcomes.*
2. *There exists a sequence of equilibria with information acquisition that has asymptotically inefficient outcomes. Along this sequence, the prior-favored policy A is elected with probability converging to one in both states.*
3. *Any equilibrium sequence with information acquisition and convergent outcome probabilities has one of two limit outcomes: either it is asymptotically efficient, or the prior-favored policy A is elected with probability converging to one in both states.*

The proof of Theorem 1 is in the appendix. It constructs the equilibrium sequences using a fixed-point theorem, established in a companion note (Ekmekci *et al.*, 2025), that extends the Poincaré–Miranda theorem.[5] Our analysis focuses on the case $q_A r_A > q_B r_B$, in which A is the favored policy given the prior. The same result extends to the case when $q_A r_A < q_B r_B$ by analogous arguments. Only in the knife-edge case $q_A r_A = q_B r_B$ studied in the main specification of Martinelli (2006), coordination is not an issue.

4.1 Information Aggregation

The inefficiency of the equilibrium sequences in Theorem 1 is not just due to voters supporting the prior-favored policy with a majority in both states. Besides that,

[5]The original Poincaré–Miranda theorem appears in Miranda (1940).

information acquisition is so low that an outside observer cannot learn the state from observing the realized vote share, as $n \to \infty$. That is, information aggregation fails in a statistical sense.

Proposition 2 *(Failure of Statistical Information Aggregation) Consider the inefficient equilibrium sequence $\alpha(x_n, \delta_n)$ constructed in the proof of Theorem 1. Let $\omega \in \{z_A, z_B\}$ denote the state, and let Y_n be the number of votes for A. After observing Y_n, let*

$$\mu_n(Y_n) = \Pr(\omega = z_A \mid Y_n).$$

Then the realized vote share does not reveal the state asymptotically. More precisely, there exists $\varepsilon > 0$ such that

$$\liminf_{n\to\infty} \Pr\left(\varepsilon \leq \mu_n(Y_n) \leq 1 - \varepsilon\right) \geq \varepsilon.$$

The proof of Proposition 2 is in the appendix. It applies the well-known Bretagnolle-Huber inequality from information theory (Bretagnolle and Huber, 1979) to the equilibrium construction from the proof of Theorem 1.

Two remarks are worth noting. First, the coordination problem can be present even in small electorates. Figure 2 below provides an example for an electorate of just 21 voters. Second, the inefficient equilibrium sequence exists even when the information costs vanish arbitrarily fast to zero as $x \to 0$. Information costs may even vanish exponentially fast; for example, the results apply to $C(x) = \exp(-1/x)$ for $x \in (0, \frac{1}{2}]$ and $C(0) = 0$. In particular, the failure of statistical information aggregation may be surprising with exponentially low cost.

The following section explains the economic mechanism that drives the voters' coordination problem and gives an intuition for the coexistence of the low-information and the high-information equilibria.

4.2 Intuition for the Coordination Problem

Economically, the coexistence of the low-information and high-information equilibria is driven by the strategic complementarities in information acquisition described in Proposition 1.

To get some intuition, it is instructive to compare the "asymmetric" setting considered here, in which A is the prior-favored policy, with the "symmetric" main setting in Martinelli (2006) in which the voters are exactly indifferent given the prior. In the symmetric setting, equilibrium incentives are governed by a pure strategic-substitute logic and no coordination problem arises, unlike in the asymmetric setting.

To explain this, we describe equilibria in both settings through a one-dimensional fixed-point equation and then illustrate the role of complementarities in an example. In the asymmetric setting, for each $x \in (0, \frac{1}{2})$ such that the indifference condition (2) has a solution, let $\delta(x)$ denote the minimal such solution. Denote by $\widehat{x}_n\big(x, \delta(x)\big)$ the precision solving the first-order condition (3) given $\alpha\Big(x, \delta(x)\Big)$. Any fixed point

$$x = \widehat{x}_n\big(x, \delta(x)\big) \tag{6}$$

of the correspondence $x \mapsto \widehat{x}_n\big(x, \delta(x)\big)$ induces a strategy $\alpha\Big(x, \delta(x)\Big)$ satisfying both equilibrium conditions (2) and (3), hence an equilibrium. In the symmetric setting, the symmetry implies that all equilibria satisfy $\delta = 0$ and are characterized by the first-order condition (3), which we rewrite as

$$x = \widehat{x}_n(x, 0); \tag{7}$$

cf. Theorem 1 in Martinelli (2006).

Figure 2 illustrates the two fixed-point equations (6) and (7) in an example. The right panel shows the mapping $x \mapsto \widehat{x}_n(x, 0)$ in the symmetric setting and the left panel shows the mapping $x \mapsto \widehat{x}_n\big(x, \delta(x)\big)$ in the asymmetric setting.

In the symmetric setting, $\delta = 0$ implies that information acquisition is locally a strategic substitute (cf. Proposition 1). As a consequence, the precision $\widehat{x}_n(x, 0)$ of the best response to $\alpha(x, 0)$ is strictly decreasing in x: for a given voter, the more information the other voters acquire, the less incentive the voter has to acquire information herself. The monotonicity implies a unique crossing point with the identity map and thus a unique equilibrium. Absent complementarities, equilibrium is unique.

In the asymmetric setting, the complementarities imply a non-monotone shape of the fixed-point mapping $x \mapsto \widehat{x}_n\big(x, \delta(x)\big)$. This enables two crossing points with the identity map, and thus two equilibria. At the high-information equilibrium, voters acquire more information relative to the low-information equilibrium, but also have stronger incentives to acquire information under the best response.

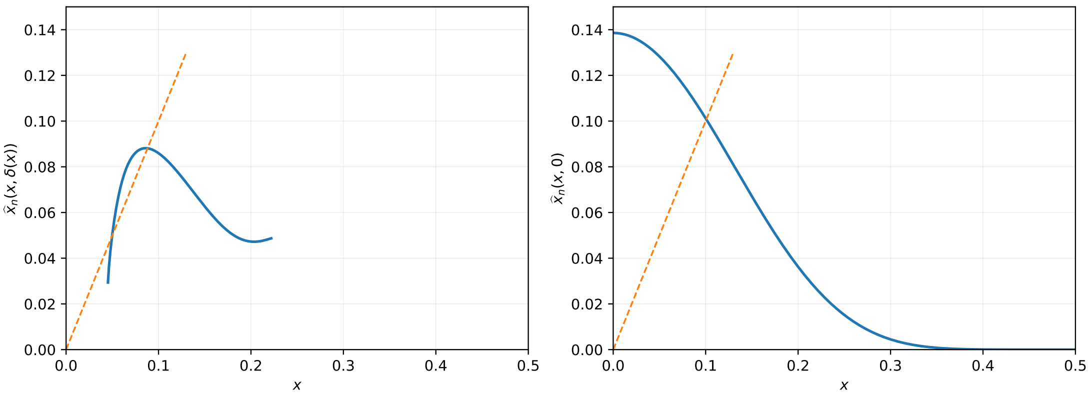

Figure 2: Example with $n = 10$, $C(x) = x^{2.3}$, and $r_A = r_B = 1$. The left panel shows the fixed-point mapping $x \mapsto \widehat{x}_n\big(x, \delta(x)\big)$ for the asymmetric setting with $q_A = 3/4$; the right panel shows the fixed-point mapping $x \mapsto \widehat{x}_n(x, 0)$ for the symmetric setting in which $q_A = 1/2$.

The asymptotic inefficiency of the low-information equilibrium can be seen through the equilibrium indifference condition (2) that ensures the voters have a best response with information acquisition. As $x \to 0$ in all equilibrium sequences, indifference between (x, A, B) and $(0, A, A)$ requires asymptotic indifference between $(0, A, B)$ and $(0, A, A)$ and thus between $(0, A, A)$ and $(0, B, B)$; hence,[6]

$$\frac{q_A r_A}{q_B r_B} \frac{\Pr(\text{piv} \mid z_A; \alpha_n)}{\Pr(\text{piv} \mid z_B; \alpha_n)} \to 1.$$

Because $q_A r_A > q_B r_B$, the pivotal event must be asymptotically more likely in state z_B than in state z_A. Using (5), we can rewrite the condition as

$$\frac{q_B r_B}{q_A r_A} = \lim_{n \to \infty} \left(\frac{\frac{1}{4} - \mathrm{MV}(z_A; x_n, \delta_n)^2}{\frac{1}{4} - \mathrm{MV}(z_B; x_n, \delta_n)^2} \right)^n.$$

Thus the indifference condition requires the election to be closer in state z_B than in state z_A.

However, this does not require B to have an expected majority in z_B. Under $\alpha(x, \delta)$,

$$\pi(z_A; \alpha(x, \delta)) - \frac{1}{2} = \frac{\delta}{2} + (1 - \delta)x, \qquad \pi(z_B; \alpha(x, \delta)) - \frac{1}{2} = \frac{\delta}{2} - (1 - \delta)x.$$

[6]The pivotal likelihood is uniformly bounded above by $\binom{2n}{n}4^{-n}$; thus the first-order condition (3) implies $x \to 0$ in all equilibrium sequences.

The high-information equilibrium has

$$\frac{\delta}{2} - (1-\delta)x < 0,$$

so B has an expected majority in state z_B. The low-information equilibrium instead has

$$0 < \frac{\delta}{2} - (1-\delta)x < \frac{\delta}{2} + (1-\delta)x.$$

In this case, the pivotal event is still more likely in z_B than in z_A, because the election is closer in z_B. This supports the indifference condition and information acquisition despite the prior bias toward A. But the expected majority is still for A in both states, preventing B from being elected with probability converging to one and therefore precluding asymptotically efficient outcomes.

4.3 Numerical Examples

Table 1 illustrates the low- and high-information equilibria for some example parameter specifications of the model.

d	n	Low-information equilibrium		High-information equilibrium	
		$\Pr(A \mid z_A)$	$\Pr(A \mid z_B)$	$\Pr(A \mid z_A)$	$\Pr(A \mid z_B)$
2.3	10	0.971402	0.889329	0.917891	0.483388
2.3	20	0.978650	0.919906	0.920809	0.444189
2.3	50	0.985112	0.945530	0.925962	0.397603
2.3	100	0.988383	0.958030	0.930093	0.365548
2.3	1000	0.994167	0.979484	0.943086	0.275069
2.3	10000	0.996658	0.988457	0.954462	0.205135
2.4	10	0.982803	0.936862	0.918069	0.387726
2.4	20	0.987578	0.955440	0.923964	0.355434
2.4	50	0.991804	0.971152	0.932049	0.313032
2.4	100	0.993885	0.978679	0.937901	0.282659
2.4	1000	0.997345	0.990934	0.954590	0.196778
2.4	10000	0.998671	0.995528	0.967423	0.133889

Table 1: Outcome probabilities at the low- and high-information equilibria for $C(x) = x^d$, $r_A = r_B = 1$, $q_A = 3/4$, and $q_B = 1/4$. The electorate has $2n+1$ voters.

A Basic Notation

Fix a symmetric strategy α used by the other $2n$ voters and write

$$\rho_A = q_A r_A, \qquad \rho_B = q_B r_B,$$

so that, by assumption $\rho_A > \rho_B$. Let

$$P_A = \Pr(\text{piv} \mid z_A; \alpha), \qquad P_B = \Pr(\text{piv} \mid z_B; \alpha),$$

and

$$R_A = P_A q_A r_A, \qquad R_B = P_B q_B r_B.$$

For a pair (x, δ), define

$$a_A(x, \delta) = \frac{\delta}{2} + (1 - \delta)x, \qquad a_B(x, \delta) = \frac{\delta}{2} - (1 - \delta)x. \tag{8}$$

Then

$$\pi(z_A; \alpha(x, \delta)) = \frac{1}{2} + a_A(x, \delta), \qquad \pi(z_B; \alpha(x, \delta)) = \frac{1}{2} + a_B(x, \delta).$$

B Proof of Lemma 1

We prove the lemma in three steps. The first one shows the "only-if" direction of the lemma's statement.

First, we show that, for all sufficiently large n, every equilibrium with information acquisition has support

$$\{(x, A, B), (0, A, A)\}$$

for some $x \in (0, \frac{1}{2})$, so it is a strategy $\alpha(x, \delta)$ for some $(x, \delta) \in (0, \frac{1}{2}) \times (0, 1)$. Second, we show that every such equilibrium satisfies the two conditions in Lemma 1, namely (2) and (3).

Third, we prove the converse "if-direction": if α is a strategy such that $\alpha = \alpha(x, \delta)$ for some $(x, \delta) \in (0, \frac{1}{2}) \times (0, 1)$ and it satisfies (2) and (3), then α is an equilibrium with information acquisition.

Step 1: The support of equilibria with information acquisition. Fix a voter and a symmetric strategy α used by the other $2n$ voters. Only pivotal events matter for payoff comparisons. Up to terms that are constant across the fixed voter's strategies, the payoff from choosing (x_i, A, B) is

$$V_I(x_i; \alpha) = (R_A + R_B)\left(\frac{1}{2} + x_i\right) - C(x_i).$$

The payoffs from the strategies $(0, A, A)$ and $(0, B, B)$ are

$$V_A(\alpha) = R_A, \qquad V_B(\alpha) = R_B.$$

For any $x_i > 0$, the strategies (x_i, A, A) and (x_i, B, B) are strictly dominated by $(0, A, A)$ and $(0, B, B)$, respectively, because they induce the same vote but add a positive information cost. The strategy (x_i, B, A) is strictly dominated by (x_i, A, B) because the signal is informative and voting with the signal raises the probability of choosing the correct policy in each state. Hence, any pure strategy with information acquisition that is used with positive probability in equilibrium must be of the form (x_i, A, B).

The function $V_I(\cdot; \alpha)$ is strictly concave because C is strictly convex. Moreover, for any strategy α with information acquisition, $R_A + R_B > 0$, and therefore

$$V_I'(0; \alpha) = R_A + R_B - C'(0) > 0.$$

Thus, there is a unique maximizer $x^*(\alpha) > 0$ of $V_I(\cdot; \alpha)$. For large enough n, this optimizer is interior, i.e. $x^*(\alpha) < \frac{1}{2}$ and given by

$$R_A + R_B = C'(x^*(\alpha)). \tag{9}$$

Indeed, the pivotal likelihood is uniformly bounded above by

$$\binom{2n}{n} 4^{-n},$$

which converges to zero. Hence, uniformly over α, and fixing any $0 < \bar{x} < \frac{1}{2}$

$$R_A + R_B \leq (q_A r_A + q_B r_B)\binom{2n}{n} 4^{-n} < C'(\bar{x})$$

for all sufficiently large n, making $x = \frac{1}{2}$ suboptimal.

Since the payoff from $(0, A, B)$ and $(0, B, A)$ is given by $\frac{1}{2}R_A + \frac{1}{2}R_B$, it is a convex combination of $V_A(\alpha)$ and $V_B(\alpha)$. If $V_A(\alpha) = V_B(\alpha)$, then $V_I(0; \alpha) = V_A(\alpha) = V_B(\alpha)$ and $V_I'(0; \alpha) > 0$, so the unique best response is $(x^*(\alpha), A, B)$. If $V_A(\alpha) \neq V_B(\alpha)$, only one of $(0, A, A)$ and $(0, B, B)$ can be a best response. Hence any equilibrium with information acquisition can have only one of the following supports:

$$\{(x^*, A, B)\}, \qquad \{(x^*, A, B), (0, A, A)\}, \qquad \{(x^*, A, B), (0, B, B)\}.$$

We now rule out the first and third possibility for all sufficiently large n when $q_A r_A > q_B r_B$.

First consider a candidate equilibrium in which all voters use (x, A, B) for some $x > 0$. Then $P_A = P_B = P$ and the interior first-order condition (9) is

$$P(q_A r_A + q_B r_B) = C'(x).$$

The strategy (x, A, B) must also yield a weakly higher payoff than $(0, A, A)$. Thus,

$$P(q_A r_A + q_B r_B)\left(\frac{1}{2} + x\right) - C(x) \geq P q_A r_A.$$

Using the first-order condition to substitute for $P(q_A r_A + q_B r_B)$ gives

$$C'(x)\left(\frac{1}{2} + x\right) - C(x) \geq C'(x)\frac{q_A r_A}{q_A r_A + q_B r_B}.$$

Since $x > 0$ and $C'(x) > 0$, division by $C'(x)$ yields

$$\frac{1}{2} + x - \frac{C(x)}{C'(x)} \geq \frac{q_A r_A}{q_A r_A + q_B r_B}.$$

Subtracting $\frac{1}{2}$ from both sides gives

$$x - \frac{C(x)}{C'(x)} \geq \frac{q_A r_A}{q_A r_A + q_B r_B} - \frac{1}{2}.$$

The right-hand side simplifies as

$$\frac{q_A r_A}{q_A r_A + q_B r_B} - \frac{1}{2} = \frac{2q_A r_A - (q_A r_A + q_B r_B)}{2(q_A r_A + q_B r_B)} = \frac{1}{2}\frac{q_A r_A - q_B r_B}{q_A r_A + q_B r_B}.$$

Rearranging thus gives

$$x - \frac{1}{2}\frac{q_A r_A - q_B r_B}{q_A r_A + q_B r_B} \geq \frac{C(x)}{C'(x)} \geq 0.$$

Thus x must be bounded away from zero. But the first-order condition and the uniform convergence of pivotal probabilities to zero imply $C'(x) \to 0$, hence $x \to 0$ as $n \to \infty$, along any equilibrium sequence. Thus (x, A, B) cannot be an equilibrium for any n sufficiently large.

Next consider a candidate equilibrium with support $\{(x, A, B), (0, B, B)\}$ for some $x > 0$. Let $\beta \in (0, 1)$ be the probability assigned to $(0, B, B)$. Then

$$\pi(z_A; \alpha) = (1 - \beta)\left(\frac{1}{2} + x\right), \qquad \pi(z_B; \alpha) = (1 - \beta)\left(\frac{1}{2} - x\right).$$

Set

$$a = (1 - \beta)\left(\frac{1}{2} + x\right), \qquad b = (1 - \beta)\left(\frac{1}{2} - x\right).$$

Then $\pi(z_A; \alpha) = a$ and $\pi(z_B; \alpha) = b$, so

$$\begin{aligned}
\pi(z_A; \alpha)(1 - \pi(z_A; \alpha)) - \pi(z_B; \alpha)(1 - \pi(z_B; \alpha)) &= a(1 - a) - b(1 - b) \\
&= (a - b)(1 - a - b) + ab - ba \\
&= (a - b)(1 - a - b).
\end{aligned}$$

Since

$$a - b = 2(1 - \beta)x \qquad \text{and} \qquad 1 - a - b = 1 - (1 - \beta) = \beta,$$

we obtain

$$\pi(z_A; \alpha)(1 - \pi(z_A; \alpha)) - \pi(z_B; \alpha)(1 - \pi(z_B; \alpha)) = 2\beta(1 - \beta)x > 0.$$

Hence $P_A > P_B$, given (1). Together with $q_A r_A > q_B r_B$, this gives

$$V_A(\alpha) = P_A q_A r_A > P_B q_B r_B = V_B(\alpha).$$

Thus $(0, B, B)$ cannot be in the support of a best response. By way of exclusion, we conclude that the support of any equilibrium with information acquisition is therefore

$$\{(x^*, A, B), (0, A, A)\}.$$

Step 2: Necessity of the two equilibrium conditions. Consider any equilibrium with information acquisition. By the first step, when n is sufficiently large, the equilibrium is a strategy $\alpha(x, \delta)$ with $(x, \delta) \in (0, \frac{1}{2}) \times (0, 1)$ and support $\{(x, A, B), (0, A, A)\}$ for some $x \in (0, \frac{1}{2})$. Since x is interior and the strategy (x, A, B) is a best response, it must satisfy the first-order condition

$$\Pr(\text{piv} \mid z_A; \alpha(x, \delta)) q_A r_A + \Pr(\text{piv} \mid z_B; \alpha(x, \delta)) q_B r_B = C'(x),$$

which is precisely (3). Since both (x, A, B) and $(0, A, A)$ are used with positive probability, they must yield the same payoff. The indifference condition is precisely (2).

Step 3: Sufficiency. It remains to prove the converse. Suppose that α is a strategy so that there is $(x, \delta) \in (0, 1/2) \times (0, 1)$ with $\alpha = \alpha(x, \delta)$ and α satisfies (2) and (3). The strategy $\alpha(x, \delta)$ has support $\{(x, A, B), (0, A, A)\}$, and (2) states that the two strategies in the support yield the same payoffs. In the following, we argue that all strategies outside the support have weakly lower payoffs. Thus, α is an equilibrium with information acquisition.

First, we show that $(0, B, B)$ is strictly worse. By (3),

$$R_A + R_B = C'(x).$$

By the indifference condition (2),

$$R_A = (R_A + R_B) \left(\frac{1}{2} + x \right) - C(x).$$

Substituting $R_A + R_B = C'(x)$ gives

$$R_A = C'(x) \left(\frac{1}{2} + x \right) - C(x).$$

Therefore,

$$R_B = C'(x) - R_A = C'(x) - C'(x) \left(\frac{1}{2} + x \right) + C(x) = C'(x) \left(\frac{1}{2} - x \right) + C(x).$$

Hence

$$\begin{aligned}
V_A(\alpha(x,\delta)) - V_B(\alpha(x,\delta)) &= R_A - R_B \\
&= \left[C'(x)\left(\frac{1}{2}+x\right) - C(x)\right] - \left[C'(x)\left(\frac{1}{2}-x\right) + C(x)\right] \\
&= 2\big(xC'(x) - C(x)\big) > 0.
\end{aligned}$$

The strict inequality follows from the strict convexity of C and $C(0) = 0$: for $x > 0$,

$$C(x) = \int_0^x C'(t)\, dt < xC'(x).$$

Thus $(0, B, B)$ is strictly worse than the two strategies in the support.

Second, the strategies $(0, A, B)$ and $(0, B, A)$ are strictly worse because their payoffs are convex combinations of the payoffs R_A and R_B from $(0, A, A)$ and $(0, B, B)$; they are given by

$$\frac{1}{2}(R_A + R_B).$$

Third, as argued above, for any $x > 0$, the strategy (x, B, A) is dominated by (x, A, B). Fourth, the payoff $V_I(y; \alpha(x, \delta))$ of any strategy (y, A, B) with $y \neq x$ yields a weakly lower payoff than (x, A, B) because $V_I(\cdot; \alpha(x, \delta))$ is strictly concave, and (3) makes x its unique maximizer.

We conclude that every strategy outside the support of $\alpha(x, \delta)$ yields weakly lower payoffs than (x, A, B) and $(0, A, A)$ given $\alpha(x, \delta)$.

C Proof of Proposition 1

For a strategy $\alpha(x, \delta)$, the marginal benefit of a higher precision is

$$\mathrm{MB}_n(x,\delta) = \binom{2n}{n}\left[\rho_A\left(\frac{1}{4} - a_A(x,\delta)^2\right)^n + \rho_B\left(\frac{1}{4} - a_B(x,\delta)^2\right)^n\right].$$

Since

$$\frac{\partial a_A(x,\delta)}{\partial x} = 1 - \delta, \qquad \frac{\partial a_B(x,\delta)}{\partial x} = -(1-\delta),$$

differentiating MB_n with respect to x gives

$$\frac{\partial \mathrm{MB}_n(x,\delta)}{\partial x} = 2n(1-\delta)\binom{2n}{n}\left[\rho_B a_B(x,\delta)\left(\frac{1}{4}-a_B(x,\delta)^2\right)^{n-1} \right.$$
$$\left. - \rho_A a_A(x,\delta)\left(\frac{1}{4}-a_A(x,\delta)^2\right)^{n-1}\right].$$

The factor outside the square brackets is strictly positive. Hence the sign of $\partial \mathrm{MB}_n(x,\delta)/\partial x$ is the sign of the expression inside the square brackets.

If $\delta = 0$, then $a_A(x,0) = x$ and $a_B(x,0) = -x$. The expression inside the square brackets becomes

$$-(\rho_A+\rho_B)x\left(\frac{1}{4}-x^2\right)^{n-1} < 0$$

for every $x \in (0, 1/2)$. Thus $\mathrm{MB}_n(\cdot, 0)$ is strictly decreasing. In the remainder of the proof, we consider $\delta > 0$.

Claim 1 *Fix $(x,\delta) \in (0,\frac{1}{2}) \times (0,1)$. If $a_B(x,\delta) \leq 0$, then*

$$\frac{\partial \mathrm{MB}_n(x,\delta)}{\partial x} < 0.$$

If $a_B(x,\delta) > 0$ and

$$s = \frac{2(1-\delta)x}{\delta},$$

then the sign of $\partial \mathrm{MB}_n(x,\delta)/\partial x$ is the sign of $\Lambda_n(s,\delta)$, where

$$\Lambda_n(s,\delta) = \log\frac{\rho_B}{\rho_A} + \log\frac{1-s}{1+s} + (n-1)\log\frac{1-\delta^2(1-s)^2}{1-\delta^2(1+s)^2}. \tag{10}$$

Proof. If $a_B(x,\delta) \leq 0$, then the first term inside the square brackets in the derivative of MB_n is weakly negative, while the second term is strictly negative since $a_A(x,\delta) = \frac{\delta}{2} + (1-\delta)x > 0$. Hence the overall derivative is strictly negative.

Now suppose $a_B(x,\delta) > 0$. Then $s < 1$, and

$$a_A(x,\delta) = \frac{\delta}{2}(1+s), \qquad a_B(x,\delta) = \frac{\delta}{2}(1-s).$$

The derivative $\partial_x \mathrm{MB}_n(x,\delta)$ is positive if and only if

$$\rho_B a_B(x,\delta)\left(\frac{1}{4}-a_B(x,\delta)^2\right)^{n-1} > \rho_A a_A(x,\delta)\left(\frac{1}{4}-a_A(x,\delta)^2\right)^{n-1}.$$

Taking logs, the inequality is equivalent to

$$\log \frac{\rho_B}{\rho_A} + \log \frac{a_B(x,\delta)}{a_A(x,\delta)} + (n-1)\log \frac{\frac{1}{4} - a_B(x,\delta)^2}{\frac{1}{4} - a_A(x,\delta)^2} > 0.$$

($x < 1/2$ implies $a_A(x,\delta) < 1/2$, so all logarithms below are well defined.) Using

$$\frac{a_B(x,\delta)}{a_A(x,\delta)} = \frac{1-s}{1+s},$$

and

$$\frac{\frac{1}{4} - a_B(x,\delta)^2}{\frac{1}{4} - a_A(x,\delta)^2} = \frac{\frac{1}{4} - \frac{\delta^2}{4}(1-s)^2}{\frac{1}{4} - \frac{\delta^2}{4}(1+s)^2} = \frac{1-\delta^2(1-s)^2}{1-\delta^2(1+s)^2},$$

the preceding inequality becomes exactly $\Lambda_n(s,\delta) > 0$. Repeating this argument with reversed signs shows that the sign of $\partial_x \mathrm{MB}_n(x,\delta)$ is the same as the sign of $\Lambda_n(s,\delta)$, as claimed. ■

Claim 2 *Fix $\delta \in (0,1)$ and let $I_\delta = \{x \in (0,\frac{1}{2}) : a_B(x,\delta) > 0\}$, $s(x) = \frac{2(1-\delta)x}{\delta}$, and $p = \frac{\delta^2}{4} - (1-\delta)^2 x^2$. For $x \in I_\delta$, the sign of $\partial\Lambda_n/\partial s$ is the sign of*

$$Q_n(p) = -(2n-1)p^2 + \frac{n-2}{2}p + \frac{\delta^2}{4} - \frac{1}{16}.$$

Moreover, for all sufficiently large n:

1. *If $0 < \delta < 1/2$, the composition $\Lambda_n(s(\cdot),\delta)$ has at most one local maximum on I_δ.*
2. *If $1/2 < \delta < 1$, the composition $\Lambda_n(s(\cdot),\delta)$ is strictly increasing on $(0,\frac{1}{2})$.*

Proof. The proof proceeds in three parts. First, we prove the claim about the sign of $\partial\Lambda_n/\partial s$. Second and third, we consider the cases $0 < \delta < \frac{1}{2}$ and $\frac{1}{2} < \delta < 1$.

Throughout the proof, we use the following shorthand notation,

$$a = a_A(x,\delta), \qquad b = a_B(x,\delta).$$

and the relations

$$\delta = a + b, \qquad p = ab. \tag{11}$$

Differentiating with respect to s or $t = \frac{\delta}{2}s$ gives the same sign. It is convenient to differentiate with respect to t. Since $a = \frac{\delta}{2} + t$ and $b = \frac{\delta}{2} - t$, we have $da/dt = 1$

and $db/dt = -1$. Using the log representation

$$\Lambda_n = \log \frac{\rho_B}{\rho_A} + \log \frac{b}{a} + (n-1) \log \frac{\frac{1}{4} - b^2}{\frac{1}{4} - a^2},$$

we obtain

$$\frac{\partial \Lambda_n}{\partial t} = -\left(\frac{1}{b} + \frac{1}{a}\right) + (n-1)\left[\frac{2b}{\frac{1}{4} - b^2} + \frac{2a}{\frac{1}{4} - a^2}\right].$$

Rearranging,

$$\frac{\partial \Lambda_n}{\partial t} = 2(n-1)\left[\frac{b}{\frac{1}{4} - b^2} + \frac{a}{\frac{1}{4} - a^2}\right] - \left(\frac{1}{b} + \frac{1}{a}\right).$$

We now simplify the two terms, using (11). First,

$$\begin{aligned}
\frac{b}{\frac{1}{4} - b^2} + \frac{a}{\frac{1}{4} - a^2} &= \frac{b(\frac{1}{4} - a^2) + a(\frac{1}{4} - b^2)}{(\frac{1}{4} - a^2)(\frac{1}{4} - b^2)} \\
&= \frac{\frac{1}{4}(a+b) - ab(a+b)}{(\frac{1}{4} - a^2)(\frac{1}{4} - b^2)} \\
&= \frac{\delta(\frac{1}{4} - p)}{(\frac{1}{4} - a^2)(\frac{1}{4} - b^2)}.
\end{aligned}$$

Second,

$$\frac{1}{b} + \frac{1}{a} = \frac{a+b}{ab} = \frac{\delta}{p}.$$

Since $\delta > 0$, $p > 0$, and $(\frac{1}{4} - a^2)(\frac{1}{4} - b^2) > 0$, multiplication of $\frac{\partial \Lambda_n}{\partial t}$ by

$$\frac{p}{\delta}\left(\frac{1}{4} - a^2\right)\left(\frac{1}{4} - b^2\right)$$

shows that the sign of $\partial \Lambda_n / \partial t$ is the sign of

$$2(n-1)p\left(\frac{1}{4} - p\right) - \left(\frac{1}{4} - a^2\right)\left(\frac{1}{4} - b^2\right).$$

We express the last product in terms of p. We have

$$\left(\frac{1}{4} - a^2\right)\left(\frac{1}{4} - b^2\right) = \frac{1}{16} - \frac{1}{4}(a^2 + b^2) + a^2 b^2.$$

Using (11), we rewrite the second and third term as

$$a^2 + b^2 = (a+b)^2 - 2ab = \delta^2 - 2p,$$

and $a^2b^2 = p^2$. Thus, the expression becomes

$$\left(\frac{1}{4} - a^2\right)\left(\frac{1}{4} - b^2\right) = \frac{1}{16} - \frac{\delta^2}{4} + \frac{p}{2} + p^2.$$

Therefore the sign of $\partial\Lambda_n/\partial t$ is the sign of

$$2(n-1)p\left(\frac{1}{4} - p\right) - \left(\frac{1}{16} - \frac{\delta^2}{4} + \frac{p}{2} + p^2\right).$$

Expanding,

$$2(n-1)p\left(\frac{1}{4} - p\right) = \frac{n-1}{2}p - 2(n-1)p^2.$$

Thus the preceding expression equals

$$Q_n(p) = -(2n-1)p^2 + \frac{n-2}{2}p + \frac{\delta^2}{4} - \frac{1}{16},$$

as claimed.

Now we turn to the case distinction.

Case 1 $0 < \delta < 1/2$

The assumption $a_B(x, \delta) > 0$ implies

$$0 < x < \frac{\delta}{2(1-\delta)}.$$

Hence

$$0 < (1-\delta)^2 x^2 < \frac{\delta^2}{4}, \qquad \text{so} \qquad p = \frac{\delta^2}{4} - (1-\delta)^2 x^2 \in \left(0, \frac{\delta^2}{4}\right).$$

Since $\delta < 1/2$, this implies $p \in (0, 1/16)$. Differentiating Q_n gives

$$Q_n'(p) = -2(2n-1)p + \frac{n-2}{2}.$$

Since $p < 1/16$,

$$Q_n'(p) > -\frac{2n-1}{8} + \frac{n-2}{2} = \frac{2n-7}{8} > 0$$

for all sufficiently large n. Hence Q_n is increasing in p. Since

$$p = \frac{\delta^2}{4} - (1-\delta)^2 x^2$$

is decreasing in x, and hence in t, the derivative $\partial\Lambda_n/\partial t$ changes sign at most once, from positive to negative. Therefore $\Lambda_n(\cdot,\delta)$ is either decreasing throughout, or else increasing and then decreasing. In particular, it has at most one local maximum. Since $t(x) = \frac{\delta}{2}s(x)$ and $s(x) = \frac{2(1-\delta)x}{\delta}$, the same is true for the composition $\Lambda_n(s(\cdot),\delta)$.

Case 2 $1/2 \le \delta < 1$

For every $x \in (0, 1/2)$,

$$a_B(x,\delta) = \frac{\delta}{2} - (1-\delta)x > \frac{\delta}{2} - \frac{1-\delta}{2} = \frac{2\delta-1}{2} \ge 0.$$

Thus $a_B(x,\delta) > 0$ for all $x \in (0, 1/2)$. The variable p ranges from

$$p_0 = \frac{\delta^2}{4} \qquad \text{as } x \downarrow 0$$

down to

$$p_1 = \frac{2\delta-1}{4} \qquad \text{as } x \uparrow \frac{1}{2}.$$

The quadratic Q_n is concave in p, so its minimum on $[p_1, p_0]$ is attained at one of the endpoints. Substituting $p_0 = \delta^2/4$ gives

$$\begin{aligned} Q_n(p_0) &= -(2n-1)\frac{\delta^4}{16} + \frac{n-2}{2}\frac{\delta^2}{4} + \frac{\delta^2}{4} - \frac{1}{16} \\ &= \frac{-(2n-1)\delta^4 + 2n\delta^2 - 1}{16}. \end{aligned}$$

Since $\delta < 1$, we see that $Q_n(p_0) > 0$ for all sufficiently large n. Similarly, substituting $p_1 = (2\delta-1)/4$ gives

$$Q_n(p_1) = -(2n-1)\frac{(2\delta-1)^2}{16} + \frac{n-2}{2}\frac{2\delta-1}{4} + \frac{\delta^2}{4} - \frac{1}{16}.$$

To simplify this expression, write $a = 2\delta - 1$. Then $a \in (0,1)$ because $\delta \in (1/2, 1)$, and

$$\frac{\delta^2}{4} - \frac{1}{16} = \frac{(2\delta-1)^2 + 2(2\delta-1)}{16} = \frac{a^2+2a}{16}.$$

Hence

$$\begin{aligned}Q_n(p_1) &= -(2n-1)\frac{a^2}{16} + \frac{n-2}{8}a + \frac{a^2+2a}{16} \\ &= \frac{-(2n-2)a^2 + 2(n-1)a}{16} \\ &= \frac{(n-1)a(1-a)}{8}.\end{aligned}$$

Returning to δ, this is

$$Q_n(p_1) = \frac{(n-1)(2\delta-1)(1-\delta)}{4}.$$

Since $\delta \in (1/2, 1)$, we have $2\delta - 1 > 0$ and $1 - \delta > 0$. Therefore

$$Q_n(p_1) > 0$$

for every $n > 1$. Together with the positivity at the other endpoint, and since Q_n is concave in p, it follows that

$$Q_n(p) > 0$$

for every p between the two endpoints, and therefore for every corresponding $x \in (0, 1/2)$. Therefore $\partial\Lambda_n/\partial t > 0$, so $\Lambda_n(\cdot, \delta)$ is strictly increasing, and so is the composition $\Lambda_n(s(\cdot), \delta)$. ∎

Claim 3 *For all sufficiently large n, there exists $\delta_{1,n} \in (0, 1/2)$ such that*

1. *For $0 < \delta < \delta_{1,n}$, $\mathrm{MB}_n(\cdot, \delta)$ is strictly decreasing for $x \in (0, 1/2)$.*
2. *For $\delta_{1,n} < \delta < 1/2$, $\mathrm{MB}_n(\cdot, \delta)$ is strictly decreasing, strictly increasing, then strictly decreasing, as x varies from 0 to $\frac{1}{2}$.*

Proof. For $0 < \delta < 1/2$, define

$$L_n(\delta) = \sup_{s\in(0,1)} \Lambda_n(s, \delta),$$

and note that

$$L_n(0) = \log \frac{\rho_B}{\rho_A}.$$

We make several observations in order to define $\delta_{1,n}$: First, L_n is weakly increasing. This is because, for each fixed $s \in (0, 1)$, the function $\Lambda_n(s, \delta)$ is increasing in

δ. Indeed, inspecting (10), only the last term in Λ_n depends on δ, and

$$\frac{\partial}{\partial\delta}\log\frac{1-\delta^2(1-s)^2}{1-\delta^2(1+s)^2} = -\frac{2\delta(1-s)^2}{1-\delta^2(1-s)^2}+\frac{2\delta(1+s)^2}{1-\delta^2(1+s)^2}$$
$$= 2\delta\left[\frac{(1+s)^2}{1-\delta^2(1+s)^2}-\frac{(1-s)^2}{1-\delta^2(1-s)^2}\right] > 0.$$

Second, $L_n(0) = \log\frac{\rho_B}{\rho_A} < 0$. Third, for every fixed $\bar{\delta} \in (0, 1/2)$, $L_n(\bar{\delta}) > 0$ if n is sufficiently large. This is because, for every fixed $s \in (0, 1)$,

$$\log\frac{1-\bar{\delta}^2(1-s)^2}{1-\bar{\delta}^2(1+s)^2} > 0,$$

so the last term in $\Lambda_n(s, \bar{\delta})$ grows linearly in n, implying the claim.

We conclude that, for all sufficiently large n, the threshold

$$\delta_{1,n} = \inf\{\delta \in [0, 1/2) : L_n(\delta) \geq 0\}$$

is well defined and belongs to $(0, 1/2)$.

Case 3 $0 < \delta < \delta_{1,n}$

For $0 < \delta < \delta_{1,n}$, we have $\Lambda_n(s, \delta) < 0$ for every $s \in (0, 1)$. If $a_B(x, \delta) > 0$, the first claim implies $\partial_x\mathrm{MB}_n(x, \delta) < 0$. If $a_B(x, \delta) \leq 0$, the first claim also implies $\partial_x\mathrm{MB}_n(x, \delta) < 0$. Therefore $\mathrm{MB}_n(\cdot, \delta)$ is strictly decreasing for $x \in (0, 1/2)$.

Case 4 $\delta_{1,n} < \delta < 1/2$

By the definition of the infimum, for every $\delta > \delta_{1,n}$ there exists $\delta_0 < \delta$ such that $L_n(\delta_0) \geq 0$. Since

$$\lim_{s\downarrow 0}\Lambda_n(s, \delta_0) = \log\frac{\rho_B}{\rho_A} < 0, \qquad \lim_{s\uparrow 1}\Lambda_n(s, \delta_0) = -\infty,$$

the supremum defining $L_n(\delta_0)$ is attained at some interior point $s_0 \in (0, 1)$. For this s_0, the map $\delta \mapsto \Lambda_n(s_0, \delta)$ is strictly increasing. Hence

$$L_n(\delta) \geq \Lambda_n(s_0, \delta) > \Lambda_n(s_0, \delta_0) = L_n(\delta_0) \geq 0.$$

Thus $L_n(\delta) > 0$. By the second claim, $\Lambda_n(\cdot, \delta)$ has at most one local maximum. Since

$$\lim_{s\downarrow 0}\Lambda_n(s, \delta) = \log\frac{\rho_B}{\rho_A} < 0, \qquad \lim_{s\uparrow 1}\Lambda_n(s, \delta) = -\infty,$$

and $L_n(\delta) > 0$, Λ_n crosses zero exactly twice in $s \in (0,1)$. By the first claim, the sign of $\partial_x \mathrm{MB}_n(x,\delta)$ is negative, then positive, then negative if $a_B(x,\delta) > 0$, i.e. for

$$x \in \left(0, \frac{\delta}{2(1-\delta)}\right).$$

For

$$x \in \left[\frac{\delta}{2(1-\delta)}, \frac{1}{2}\right),$$

we have $a_B(x,\delta) \leq 0$, so the first claim implies $\partial_x \mathrm{MB}_n(x,\delta) < 0$. We conclude that $\mathrm{MB}_n(\cdot,\delta)$ is strictly decreasing, then strictly increasing, and then strictly decreasing as x varies from 0 to $\frac{1}{2}$. ■

Claim 4 *If $1/2 < \delta < 1$, then $\mathrm{MB}_n(\cdot,\delta)$ is strictly decreasing and then strictly increasing for $x \in (0, 1/2)$.*

Proof. Fix $1/2 < \delta < 1$. As shown above,

$$a_B(x,\delta) > 0 \qquad \text{for every } x \in (0, 1/2).$$

Hence, by the first claim, the sign of $\partial_x \mathrm{MB}_n(x,\delta)$ is the sign of $\Lambda_n(s,\delta)$, where

$$s = \frac{2(1-\delta)x}{\delta}.$$

For $x \in (0, \frac{1}{2})$, this variable satisfies

$$s \in (0, \frac{1-\delta}{\delta}).$$

We make three observations: First, by the second claim, $\Lambda_n(\cdot,\delta)$ is strictly increasing if n is sufficiently large. Second,

$$\lim_{s \downarrow 0} \Lambda_n(s,\delta) = \log \frac{\rho_B}{\rho_A} < 0.$$

At the other endpoint, as $s \uparrow \frac{(1-\delta)}{\delta}$, $1-\delta^2(1+s)^2 \downarrow 0$. Therefore the last logarithmic term in $\Lambda_n(s,\delta)$ diverges to $+\infty$. Since $\delta > 1/2$, we have $\frac{1-\delta}{\delta} < 1$, thus $s < 1$ and the second term $\log \frac{1-s}{1+s}$ remains finite at the upper endpoint. Hence, for all sufficiently large n,

$$\lim_{s \uparrow (1-\delta)/\delta} \Lambda_n(s,\delta) = +\infty.$$

To summarize the three observations: $\Lambda_n(\cdot,\delta)$ is strictly increasing, starts below zero, and tends to $+\infty$. Thus, it crosses zero exactly once. Therefore $\partial_x \mathrm{MB}_n(x, \delta)$ is negative before this crossing and positive after it. Hence $\mathrm{MB}_n(\cdot, \delta)$ is strictly decreasing and then strictly increasing as x varies from 0 to $\frac{1}{2}$. ■

Combining the preliminary case $\delta = 0$ with Claims 3 and 4 proves Proposition 1.

D Mathematical Preliminaries for the Proof of Theorem 1

Central limit approximation. Consider a sequence of strategies $(\alpha_n)_{n\in\mathbb{N}}$, and write

$$\pi_n(z) = \pi(z; \alpha_n), \qquad N_n = 2n + 1.$$

For the sequences considered below, assume $0 < \pi_n(z) < 1$ and define

$$s_n(z; \alpha_n) = \left(\frac{N_n}{\pi_n(z)(1 - \pi_n(z))} \right)^{-1/2}, \qquad \Delta_n(z; \alpha_n) = \frac{\pi_n(z) - \frac{n}{N_n}}{s_n(z; \alpha_n)}.$$

Thus $s_n(z; \alpha_n)$ is the standard deviation of the vote share for A in state z, and $\Delta_n(z; \alpha_n)$ is the distance between the expected vote share and the majority threshold, measured in standard deviations.

Let $X_n(z) \sim \mathrm{Bin}(N_n, \pi_n(z))$ denote the number of votes for A in state z, and set

$$Z_n(z) = \frac{X_n(z) - N_n \pi_n(z)}{\sqrt{N_n \pi_n(z)(1 - \pi_n(z))}}.$$

Since A is elected if and only if $X_n(z) > n$,

$$\Pr(A \text{ is elected} \mid z; \alpha_n) = \Pr\big(Z_n(z) > -\Delta_n(z; \alpha_n)\big).$$

The proof of Theorem 1 uses the following implication: whenever $\Delta_n(z; \alpha_n)$ converges in $\mathbb{R} \cup \{-\infty, +\infty\}$,

$$\lim_{n\to\infty} \Pr(A \text{ is elected} \mid z; \alpha_n) = \lim_{n\to\infty} \Phi\big(\Delta_n(z; \alpha_n)\big), \tag{12}$$

where Φ is the standard normal distribution function, with $\Phi(-\infty) = 0$ and $\Phi(+\infty) = 1$.

To see (12), first suppose that

$$N_n \pi_n(z)(1 - \pi_n(z)) \to \infty.$$

Then the Berry–Esseen bound for Bernoulli triangular arrays implies

$$\sup_{t \in \mathbb{R}} |\Pr(Z_n(z) \le t) - \Phi(t)| \to 0;$$

see, e.g., Feller (1971, Theorem XVI.5.1). Hence[7]

$$\Pr(A \text{ is elected} \mid z; \alpha_n) = 1 - \Pr\big(Z_n(z) \le -\Delta_n(z; \alpha_n)\big) = \Phi\big(\Delta_n(z; \alpha_n)\big) + o(1).$$

It remains to consider subsequences along which the variance does not diverge. Any such subsequence has a further subsequence along which $N_n \pi_n(z)(1 - \pi_n(z))$ is bounded. Along this further subsequence,

$$\pi_n(z)(1 - \pi_n(z)) \to 0,$$

so, after passing to a further subsequence if necessary, either $\pi_n(z) \to 0$ or $\pi_n(z) \to 1$. If $\pi_n(z) \to 0$, then $\Delta_n(z; \alpha_n) \to -\infty$ and $\Phi(\Delta_n(z; \alpha_n)) \to 0$. Further,

$$\Pr(A \text{ is elected} \mid z; \alpha_n) = \Pr\left(\frac{X_n(z)}{N_n} > \frac{n}{N_n}\right) \le \frac{N_n \pi_n(z)}{n} \to 0,$$

by Markov's inequality. If $\pi_n(z) \to 1$, then $\Delta_n(z; \alpha_n) \to +\infty$, and similarly

$$\Pr(A \text{ is not elected} \mid z; \alpha_n) = \Pr\left(N_n - X_n(z) \ge n + 1\right) \le \frac{N_n(1 - \pi_n(z))}{n + 1} \to 0.$$

Thus every subsequential limit of the left-hand side of (12) agrees with the corresponding limit of $\Phi(\Delta_n(z; \alpha_n))$, proving (12).

Stirling approximation. We will repeatedly use the following approximation of the pivotal likelihood in (1), based on Stirling's formula:

$$\Pr(\text{piv} \mid z; \alpha_n) = \frac{1 + o(1)}{\sqrt{\pi n}} \Big(4\pi(z; \alpha_n)(1 - \pi(z; \alpha_n))\Big)^n. \tag{13}$$

[7] A term $o(1)$ denotes any quantity that converges to zero in the relevant limit; for example, $r_n = o(1)$ means $\lim_{n \to \infty} r_n = 0$.

Thus, the pivotal likelihood is exponentially small unless $\pi(z;\alpha_n)$ is within order $n^{-1/2}$ of $1/2$. Based on the above approximation, one obtains

$$\begin{aligned}\Pr(\text{piv} \mid z_A;\alpha_n) &= \frac{1+o(1)}{\sqrt{\pi n}}\Big(4(\frac{1}{2}+a_{A,n})(\frac{1}{2}-a_{A,n})\Big)^n \\ &= \frac{1+o(1)}{\sqrt{\pi n}}(1-4a_{A,n}^2)^n.\end{aligned}$$

where the second equality uses the differences of squares identity. Consequently,

$$\frac{\Pr(\text{piv} \mid z_A;\alpha_n)}{n^{-\frac{1}{2}}} \to \frac{1}{\sqrt{\pi}}e^{-y} \tag{14}$$

for

$$y_n = 4na_{A,n}^2$$

since $\lim_{n\to\infty}(1-\frac{y_n}{n})^n = \lim_{n\to\infty} e^{-y_n}$ for $y_n \in \mathbb{R}$ and since the exponential function is continuous.

Uniform Bound for Pivotal Likelihood. We repeatedly use a uniform bound for the pivotal likelihood. Namely,

$$\Pr(\text{piv} \mid z;\alpha) = \binom{2n}{n}\Big(\pi(z;\alpha)(1-\pi(z;\alpha))\Big)^n \leq \binom{2n}{n}4^{-n}. \tag{15}$$

This holds because the function $\pi(1-\pi)$ has its unique maximum at $\pi = \frac{1}{2}$.

A Generalized Poincaré–Miranda theorem. We use the following immediate corollary of Ekmekci *et al.* (2025, Theorem 1).

Lemma 2 *Let $D = [\underline{x},\overline{x}] \times [\underline{\delta},\overline{\delta}]$, and let $F, G : D \to \mathbb{R}$ be continuous. Suppose that*

$$F(\underline{x},\delta) < 0 < F(\overline{x},\delta) \qquad \textit{for all } \delta \in [\underline{\delta},\overline{\delta}],$$

and that

$$G(x,\underline{\delta}) > 0 \quad \textit{whenever } F(x,\underline{\delta}) = 0, \qquad G(x,\overline{\delta}) < 0 \quad \textit{whenever } F(x,\overline{\delta}) = 0.$$

Then there exists $(x^*,\delta^*) \in \operatorname{int} D$ such that

$$F(x^*,\delta^*) = G(x^*,\delta^*) = 0.$$

Proof. Map $[0,1]^2$ onto D by

$$T(r,t) = \big(\underline{x} + t(\overline{x} - \underline{x}), \underline{\delta} + r(\overline{\delta} - \underline{\delta})\big).$$

Apply Ekmekci *et al.* (2025, Theorem 1) to

$$f(r,t) = \frac{F(T(r,t))}{1 + |F(T(r,t))|}, \qquad g(r,t) = \frac{G(T(r,t))}{1 + |G(T(r,t))|}.$$

This normalization only restricts the range to $[-1,1]$ and preserves zeros. Theorem 1 in Ekmekci *et al.* (2025) gives a common interior zero of f and g, hence an interior common zero of F and G. ■

E Proof of Theorem 1

The first part of the theorem is the existence result of Martinelli (2006); see Theorem 4 therein. Here, we first prove the third part and then the second part.

One more preliminary observation: the assumption $C''(0) = 0$ implies

$$C'(y) = o(y) \quad \text{and} \quad C(y) = o(y^2) \qquad \text{as } y \downarrow 0. \tag{16}$$

E.1 Third part of Theorem 1

Consider any sequence of equilibria $\alpha(x_n, \delta_n)$ with information acquisition. Write

$$a_{A,n} = a_A(x_n, \delta_n) = \frac{\delta_n}{2} + (1 - \delta_n)x_n, \qquad a_{B,n} = a_B(x_n, \delta_n) = \frac{\delta_n}{2} - (1 - \delta_n)x_n.$$

We first claim that

$$\sqrt{n}\, a_{A,n} \to \infty.$$

Suppose not. Then, $a_{A,n} = O(n^{-1/2})$. By the approximation (13), $P_A(x_n, \delta_n) = \Pr(\text{piv}|z_A; \alpha(x_n, \delta_n))$ is bounded below by a positive multiple of $n^{-1/2}$. Hence

$$\text{MB}_n(x_n, \delta_n) = R_A(x_n, \delta_n) + R_B(x_n, \delta_n) \geq kn^{-1/2} \tag{17}$$

for some $k > 0$. Since both terms in $a_{A,n} = \frac{\delta_n}{2} + (1 - \delta_n)x_n$ are nonnegative, $a_{A,n} = O(n^{-1/2})$ implies $x_n = O(n^{-1/2})$. Combining this with (16), we see that

$$C'(x_n) \leq C'(Kn^{-1/2}) = o(n^{-1/2}) \tag{18}$$

for some $K > 0$. But the first-order condition demands

$$C'(x_n) = \mathrm{MB}_n(x_n, \delta_n), \tag{19}$$

Jointly, (17)-(19) imply a contradiction. Thus $\sqrt{n}\, a_{A,n} \to \infty$.

Next we claim that

$$\sqrt{n}\, |a_{B,n}| \to \infty.$$

Suppose not. Then, $|a_{B,n}| = O(n^{-1/2})$. By the approximation (13), $P_B(x_n, \delta_n) := \Pr(\text{piv}|z_B; \alpha(x_n, \delta_n))$ is bounded below by a positive multiple of $n^{-1/2}$.

We now use the equilibrium indifference condition (2). Since

$$\mathrm{MB}_n(x_n, \delta_n) = R_A(x_n, \delta_n) + R_B(x_n, \delta_n),$$

the indifference condition can be written as

$$\big(R_A(x_n, \delta_n) + R_B(x_n, \delta_n)\big)\left(\frac{1}{2} + x_n\right) - C(x_n) = R_A(x_n, \delta_n).$$

Dividing by $R_A(x_n, \delta_n) + R_B(x_n, \delta_n)$ gives

$$\frac{R_A(x_n, \delta_n)}{R_A(x_n, \delta_n) + R_B(x_n, \delta_n)} = \frac{1}{2} + x_n - \frac{C(x_n)}{R_A(x_n, \delta_n) + R_B(x_n, \delta_n)}.$$

Using the first-order condition

$$R_A(x_n, \delta_n) + R_B(x_n, \delta_n) = C'(x_n),$$

we obtain

$$\frac{R_A(x_n, \delta_n)}{R_A(x_n, \delta_n) + R_B(x_n, \delta_n)} = \frac{1}{2} + x_n - \frac{C(x_n)}{C'(x_n)}.$$

By convexity and $C(0) = 0$, we have $C(x_n) \leq x_n C'(x_n)$, and hence the right-hand side is at least $1/2$. Therefore

$$R_A(x_n, \delta_n) \geq R_B(x_n, \delta_n).$$

Since $R_z = \rho_z P_z$, the lower bound on $P_B(x_n, \delta_n)$ implies that $P_A(x_n, \delta_n)$ is also bounded below by a positive multiple of $n^{-1/2}$. By the approximation (13), this implies $a_{A,n} = O(n^{-1/2})$, contradicting the first claim. Thus $\sqrt{n}\, |a_{B,n}| \to \infty$.

By the central limit theorem (its implication (12)), $\sqrt{n}\, a_{A,n} \to \infty$ implies that A is elected in state z_A with probability converging to one. In z_B, every subsequence has a further subsequence along which either $a_{B,n} > 0$ or $a_{B,n} < 0$. In the first case, $\sqrt{n}\, a_{B,n} \to \infty$, so A is elected in z_B with probability converging to one. In the second case, $\sqrt{n}\, a_{B,n} \to -\infty$, so B is elected in z_B with probability converging to one. Therefore, if the original sequence has convergent outcome probabilities, only one of the following two limiting outcomes can occur: either the sequence is asymptotically efficient, or A is elected with probability converging to one in both states.

E.2 Second part of Theorem 1

We prove the second part by constructing a sequence of equilibria α_n with information acquisition and asymptotically inefficient outcomes. Throughout, we use the notation $\widehat{x}_n$ for the unique precision that maximizes the expected utility across all strategies (x, A, B), given a candidate strategy α_n. Equivalently, $\widehat{x}_n$ solves the first-order condition

$$\mathrm{MB}_n(x_n, \delta_n) = C'(\widehat{x}_n).$$

The proof proceeds in three steps. First, we define a domain D_n of candidate strategies and relate the two equilibrium conditions (2) and (3) in Lemma 1 to roots of two functions F_n and G_n. Second, we establish the boundary conditions of the generalized Poincaré–Miranda theorem (Lemma 2) applied to F_n and G_n on D_n. Third, we use the Poincaré–Miranda theorem to construct an equilibrium sequence in which A is elected with probability converging to one in both states.

Step 1: The Domain of Candidate Strategies and the Equilibrium Conditions. We consider candidate equilibrium strategies α_n, given by pairs $(x_n, \delta_n) \in D_n$ from the domain

$$D_n = \{(x, \delta) : x \in \left[0, Mn^{-\frac{1}{2}}\right], \delta \in \left[3Mn^{-\frac{1}{2}}, \overline{\delta}_n\right]\} \tag{20}$$

for some $M > 0$ and a $\overline{\delta}_n$ close to 1 that we will define in the course of the proof.

For $(x, \delta) \in D_n$, define

$$F_n(x, \delta) = \max_{\tilde{x} \in [0,1/2]} \left\{ \mathrm{MB}_n(x, \delta) \left(\frac{1}{2} + \tilde{x} \right) - C(\tilde{x}) - R_A(x, \delta) \right\},$$

and

$$G_n(x,\delta) = \mathrm{MB}_n(x,\delta) - C'(x).$$

At any interior root (x,δ) with $F_n(x,\delta) = G_n(x,\delta) = 0$, the equation $G_n(x,\delta) = 0$ is exactly the first-order condition (3). Since the maximand in the definition of F_n is strictly concave in $\tilde{x}$, the first-order condition (3) implies that $\tilde{x} = x$ is its unique maximizer. Hence $F_n(x,\delta) = 0$ becomes $\mathrm{MB}_n(x,\delta)\left(\frac{1}{2} + x\right) - C(x) - R_A(x,\delta) = 0$, which is precisely the indifference condition (2) between (x, A, B) and $(0, A, A)$. Thus the two equations $F_n = 0$ and $G_n = 0$ imply the two equilibrium conditions in Lemma 1.

Step 2: The boundary conditions of the generalized Poincaré–Miranda theorem.

Boundary $x = 0$. Let $x = 0$. Then $\pi(z_A; \alpha(0,\delta)) = \pi(z_B; \alpha(0,\delta))$ and $\Pr(\mathrm{piv}|z_A; \alpha(0,\delta)) = \Pr(\mathrm{piv}|z_B; \alpha(0,\delta))$. Set $P_n(\delta) = \Pr(\mathrm{piv}|z_A; \alpha(0,\delta_n))$. Since

$$P_n(\delta) \le \binom{2n}{n} 4^{-n} = O(n^{-1/2}),$$

the maximizer in the definition of $F_n(0,\delta)$ converges uniformly to 0 for all δ. Hence, uniformly,

$$F_n(0,\delta) \le P_n(\delta)\left[-\frac{\rho_A - \rho_B}{2} + o(1)\right] < 0$$

for all sufficiently large n.

Intuitively, since all voters strictly prefer A given the prior belief, for n sufficiently large, $\widehat{x}_n$ is so small that, given $\widehat{x}_n$, after any signal $s \in \{s_A, s_B\}$, the voter strictly prefers A over B. Thus, $(\widehat{x}_n, A, A)$ yields strictly more utility than $(\widehat{x}_n, A, B)$. Further, $(0, A, A)$ yields strictly more utility than $(\widehat{x}_n, A, A)$ since the voter does not acquire costly information. Thus, all voters strictly prefer $(0, A, A)$ to $(\widehat{x}_n, A, B)$, as captured by the inequality $F_n(0,\delta) < 0$.

Boundary $x = Mn^{-\frac{1}{2}}$. Choose $M > 0$ so large that

$$\frac{\rho_A}{\rho_B} e^{-8M^2} < 1. \tag{21}$$

We show that this implies $F_n(Mn^{-1/2}, \delta) > 0$ for all $\delta \geq 3Mn^{-1/2}$ and all sufficiently large n. Let $x = Mn^{-1/2}$. Following the same algebra as for the Stirling approximation and since the common binomial factor cancels out,

$$\frac{P_A(x,\delta)}{P_B(x,\delta)} = \left(\frac{1-4a_A(x,\delta)^2}{1-4a_B(x,\delta)^2}\right)^n.$$

Since

$$\begin{aligned} a_A(x,\delta)^2 - a_B(x,\delta)^2 &= \big(a_A(x,\delta) - a_B(x,\delta)\big)\big(a_A(x,\delta) + a_B(x,\delta)\big) \\ &= (2(1-\delta)x)\,\delta \\ &= 2\delta(1-\delta)x. \end{aligned} \tag{22}$$

we can rewrite this as

$$\frac{P_A(x,\delta)}{P_B(x,\delta)} = \left(1 - \frac{a_A(x,\delta)^2 - a_B(x,\delta)^2}{\frac{1}{4} - a_B(x,\delta)^2}\right)^n = \left(1 - \frac{2\delta(1-\delta)x}{\frac{1}{4} - a_B(x,\delta)^2}\right)^n. \tag{23}$$

Moreover,

$$\frac{1}{4} - a_B(x,\delta)^2 = \left(\frac{1}{2} - a_B(x,\delta)\right)\left(\frac{1}{2} + a_B(x,\delta)\right).$$

Using (8), we rewrite

$$\frac{1}{2} - a_B(x,\delta) = (1-\delta)\left(\frac{1}{2} + x\right).$$

Since $\frac{1}{2} + a_B(x,\delta) \leq 1$, we obtain the bound

$$\frac{1}{4} - a_B(x,\delta)^2 \leq (1-\delta)\left(\frac{1}{2} + x\right).$$

Thus

$$\frac{P_A(x,\delta)}{P_B(x,\delta)} \leq \left(1 - \frac{2\delta x}{\frac{1}{2} + x}\right)^n. \tag{24}$$

For $x = Mn^{-1/2}$ and $\delta \geq 3Mn^{-1/2}$, we have $\frac{2\delta x}{\frac{1}{2}+x} \geq \frac{6M^2/n}{\frac{1}{2}+Mn^{-1/2}}$. Since, for all n sufficiently large,

$$\frac{1}{2} + Mn^{-1/2} \leq \frac{3}{4},$$

it holds

$$\frac{2\delta x}{\frac{1}{2} + x} \geq \frac{8M^2}{n},$$

so that the term on the right side of (24) is at most $(1-8M^2/n)^n$. Using $\lim_{n\to\infty}(1-\frac{y}{n})^n = e^{-y}$,

$$\frac{P_A(x,\delta)}{P_B(x,\delta)} \le e^{-8M^2}.$$

By (21), this implies

$$R_A(x,\delta) < R_B(x,\delta).$$

Evaluating the maximand in the definition of F_n at $\tilde{x} = 0$ therefore gives

$$F_n(x,\delta) \ge \frac{R_A(x,\delta) + R_B(x,\delta)}{2} - R_A(x,\delta) = \frac{R_B(x,\delta) - R_A(x,\delta)}{2} > 0.$$

Therefore $F_n(x,\delta) > 0$ for $x = Mn^{-1/2}$.

Boundary $\delta = 3Mn^{-\frac{1}{2}}$. Let $\delta = 3Mn^{-1/2}$. It follows from (8) that, for $x \in [0, Mn^{-1/2}]$, the terms $4na_A(x,\delta)^2$ and $4na_B(x,\delta)^2$ are bounded above and below by positive constants, uniformly in x. Thus, (14) implies that there is $\bar{n}$ and a positive constant so that $\mathrm{MB}_n(x,\delta)/n^{-1/2}$ exceeds the constant for all $n \ge \bar{n}$ and all x. By (16) and convexity of C,

$$C'(x) \le C'(Mn^{-1/2}) = o(n^{-1/2}).$$

We conclude that

$$G_n(x, 3Mn^{-1/2}) = \mathrm{MB}_n(x, 3Mn^{-1/2}) - C'(x) > 0$$

for $n \ge \bar{n}$.

Boundary $\delta = \overline{\delta}_n$. Choose $\kappa > 0$ so small that

$$\frac{\rho_B}{\rho_A} e^{17\kappa} < 1. \tag{25}$$

For each sufficiently large n, choose $\overline{\delta}_n \in (3Mn^{-1/2}, 1)$ sufficiently close to one such that $\overline{\delta}_n \ge 1/2$ and

$$\sup_{x\in[0,Mn^{-1/2}]} \mathrm{MB}_n(x, \overline{\delta}_n) < C'(\kappa/n). \tag{26}$$

Such a choice is possible because, for fixed n, the pivotal probabilities converge uniformly to zero on $[0, Mn^{-1/2}]$ as $\delta \to 1$.

We verify the required boundary condition. Let $x \in [0, Mn^{-1/2}]$ satisfy $F_n(x, \overline{\delta}_n) = 0$. We first claim that $x > 2\kappa/n$. Suppose instead that $x \leq 2\kappa/n$.

Applying steps analogous to the derivation of (23) gives

$$\frac{P_B(x, \overline{\delta}_n)}{P_A(x, \overline{\delta}_n)} = \left(1 + \frac{a_A(x, \overline{\delta}_n)^2 - a_B(x, \overline{\delta}_n)^2}{\frac{1}{4} - a_A(x, \overline{\delta}_n)^2}\right)^n.$$

Moreover,

$$\frac{1}{4} - a_A(x, \overline{\delta}_n)^2 = \left(\frac{1}{2} - a_A(x, \overline{\delta}_n)\right)\left(\frac{1}{2} + a_A(x, \overline{\delta}_n)\right),$$

and, by (8),

$$\frac{1}{2} - a_A(x, \overline{\delta}_n) = (1 - \overline{\delta}_n)\left(\frac{1}{2} - x\right).$$

Combining these observations with (22),

$$\frac{a_A(x, \overline{\delta}_n)^2 - a_B(x, \overline{\delta}_n)^2}{\frac{1}{4} - a_A(x, \overline{\delta}_n)^2} = \frac{2\overline{\delta}_n x}{\left(\frac{1}{2} - x\right)\left(\frac{1}{2} + a_A(x, \overline{\delta}_n)\right)}. \tag{27}$$

If $x \leq 2\kappa/n$, then the numerator is at most $4\kappa/n$. Moreover, since $a_A(x, \overline{\delta}_n) \geq 0$ and $x \to 0$, the denominator is bounded below by a number arbitrarily close to $1/4$ when n is arbitrarily large. Hence, for all sufficiently large n, the right side of (27) is bounded above by $17\kappa/n$. Therefore

$$\lim_{n\to\infty} \frac{P_B(x, \overline{\delta}_n)}{P_A(x, \overline{\delta}_n)} \leq \lim_{n\to\infty}\left(1 + \frac{17\kappa}{n}\right)^n = e^{17\kappa}.$$

Hence, by (25), there exists $\eta < 1$, such that, for all sufficiently large n,

$$R_B(x, \overline{\delta}_n) \leq \eta R_A(x, \overline{\delta}_n).$$

Let $\widehat{x}_n$ denote the maximizer in the definition of $F_n(x, \overline{\delta}_n)$. By (26), the objective in the definition of F_n has negative derivative at $\tilde{x} = \kappa/n$. Since this objective is strictly concave in $\tilde{x}$, its maximizer $\widehat{x}_n$ satisfies $\widehat{x}_n \leq \kappa/n$. Using $\mathrm{MB}_n = R_A + R_B$, we get

$$\begin{aligned}
F_n(x,\overline{\delta}_n) &= \mathrm{MB}_n(x,\overline{\delta}_n)\left(\frac{1}{2}+\widehat{x}_n\right) - C(\widehat{x}_n) - R_A(x,\overline{\delta}_n) \\
&= -\frac{R_A(x,\overline{\delta}_n) - R_B(x,\overline{\delta}_n)}{2} + \mathrm{MB}_n(x,\overline{\delta}_n)\widehat{x}_n - C(\widehat{x}_n) \\
&\leq -\frac{R_A(x,\overline{\delta}_n) - R_B(x,\overline{\delta}_n)}{2} + (R_A(x,\overline{\delta}_n) + R_B(x,\overline{\delta}_n))\frac{\kappa}{n}.
\end{aligned}$$

Since $R_B(x,\overline{\delta}_n) \leq \eta R_A(x,\overline{\delta}_n)$ for some $\eta < 1$,we have $\frac{R_A(x,\overline{\delta}_n)-R_B(x,\overline{\delta}_n)}{R_A(x,\overline{\delta}_n)+R_B(x,\overline{\delta}_n)} \geq \frac{1-\eta}{1+\eta} > 0$. Therefore the last display is strictly negative for all sufficiently large n, contradicting $F_n(x,\overline{\delta}_n) = 0$. We conclude that $F_n(x,\overline{\delta}_n) = 0$ implies $x > 2\kappa/n$.

Therefore,

$$\mathrm{MB}_n(x,\overline{\delta}_n) \leq \sup_{x' \in [0, Mn^{-1/2}]} \mathrm{MB}_n(x',\overline{\delta}_n) < C'(\kappa/n) < C'(x),$$

where the second inequality restates (26) and the third follows from $x > 2\kappa/n$ and the strict monotonicity of C'. We conclude that

$$G_n(x,\overline{\delta}_n) = \mathrm{MB}_n(x,\overline{\delta}_n) - C'(x) < 0.$$

Step 3: Construction of an inefficient equilibrium sequence. We construct an equilibrium sequence in which A is elected with probability converging to one in both states.

Applying the generalized Poincaré–Miranda theorem (Lemma 2) to F_n and G_n on the domain D_n yields, for every sufficiently large n, an interior point

$$(x_n, \delta_n) \in D_n$$

such that

$$F_n(x_n,\delta_n) = G_n(x_n,\delta_n) = 0.$$

By the discussion above, these two equations imply the two equilibrium conditions in Lemma 1. Hence $\alpha(x_n,\delta_n)$ is a symmetric equilibrium with information acquisition.

Since the roots lie in the interior of D_n, we have

$$0 < x_n < Mn^{-1/2} \qquad \text{and} \qquad \delta_n > 3Mn^{-1/2}.$$

Thus $\delta_n > 2x_n$. Using (4),

$$\pi(z_B; \alpha(x_n, \delta_n)) = \frac{1}{2} + \frac{\delta_n}{2} - (1 - \delta_n)x_n > \frac{1}{2}.$$

The sequence of outcome-probability pairs

$$(\Pr(A \text{ is elected} \mid z_A; \alpha(x_n, \delta_n)), \Pr(A \text{ is elected} \mid z_B; \alpha(x_n, \delta_n)))$$

is contained in the compact set $[0, 1]^2$. Consider any convergent subsequence. Since $\pi(z_B; \alpha(x_n, \delta_n)) > 1/2$, the probability that B is elected in state z_B cannot converge to one along that subsequence. Hence the subsequential limit cannot be asymptotically efficient. By the third part of Theorem 1, the only possible limit outcome is therefore that A is elected with probability one in both states. Since every convergent subsequence has this same limit, the full sequence of equilibria $\alpha(x_n, \delta_n)$ has outcome probabilities converging to $(1, 1)$. In particular, the outcomes are asymptotically inefficient.

E.3 Proof of Proposition 2

We use the shorthand notation

$$\pi_{A,n} = \pi(z_A; \alpha(x_n, \delta_n)), \qquad \pi_{B,n} = \pi(z_B; \alpha(x_n, \delta_n)).$$

Conditional on z_A, $Y_n \sim \mathrm{Bin}(2n + 1, \pi_{A,n})$, while conditional on z_B, $Y_n \sim \mathrm{Bin}(2n + 1, \pi_{B,n})$. Let $Q_{A,n}$ and $Q_{B,n}$ denote these two distributions.

Along the equilibrium sequence constructed in the proof of Theorem 1, for all sufficiently large n,

$$0 < x_n < Mn^{-1/2}, \qquad \delta_n > 3Mn^{-1/2}.$$

Hence $\delta_n > 2x_n$, and therefore

$$\pi_{B,n} = \frac{1}{2} + \frac{\delta_n}{2} - (1 - \delta_n)x_n > \frac{1}{2}.$$

Moreover,

$$\pi_{A,n} - \pi_{B,n} = 2(1 - \delta_n)x_n, \qquad 1 - \pi_{B,n} = (1 - \delta_n)\left(\frac{1}{2} + x_n\right).$$

We first bound the relative entropy $D(Q_{A,n}\|Q_{B,n})$ of $Q_{A,n}$ with respect to $Q_{B,n}$:

$$D(Q_{A,n}\|Q_{B,n}) = \sum_y Q_{A,n}(y)\log\frac{Q_{A,n}(y)}{Q_{B,n}(y)}.$$

Since $Q_{A,n} = \mathrm{Bin}(2n+1, \pi_{A,n})$ and $Q_{B,n} = \mathrm{Bin}(2n+1, \pi_{B,n})$, this equals

$$D(Q_{A,n}\|Q_{B,n}) = (2n+1)\left[\pi_{A,n}\log\frac{\pi_{A,n}}{\pi_{B,n}} + (1-\pi_{A,n})\log\frac{1-\pi_{A,n}}{1-\pi_{B,n}}\right].$$

For Bernoulli probabilities $p, q \in (0,1)$, the elementary inequality

$$\log t \le t - 1 \qquad \text{for all } t > 0$$

implies

$$\begin{aligned}
p\log\frac{p}{q} + (1-p)\log\frac{1-p}{1-q} &\le p\left(\frac{p}{q} - 1\right) + (1-p)\left(\frac{1-p}{1-q} - 1\right)\\
&= \frac{p(p-q)}{q} + \frac{(1-p)(q-p)}{1-q}\\
&= (p-q)\left(\frac{p}{q} - \frac{1-p}{1-q}\right)\\
&= (p-q)\frac{p(1-q) - q(1-p)}{q(1-q)}\\
&= (p-q)\frac{p-q}{q(1-q)}\\
&= \frac{(p-q)^2}{q(1-q)}.
\end{aligned}$$

Thus

$$\begin{aligned}
D(Q_{A,n}\|Q_{B,n}) &\le (2n+1)\frac{(\pi_{A,n} - \pi_{B,n})^2}{\pi_{B,n}(1-\pi_{B,n})}\\
&= (2n+1)\frac{4(1-\delta_n)^2 x_n^2}{\pi_{B,n}(1-\delta_n)(\frac{1}{2} + x_n)}\\
&\le 16(2n+1)x_n^2 \le 48M^2,
\end{aligned}$$

where the penultimate inequality uses $\pi_{B,n} > 1/2$, $\frac{1}{2} + x_n > 1/2$, and $1 - \delta_n \le 1$.

By the Bretagnolle-Huber inequality (Bretagnolle and Huber, 1979),

$$\sum_y \min\{Q_{A,n}(y), Q_{B,n}(y)\} \ge \frac{1}{2}\exp\{-D(Q_{A,n}\|Q_{B,n})\}.$$

Therefore

$$\sum_y \min\{Q_{A,n}(y), Q_{B,n}(y)\} \geq \frac{1}{2}e^{-48M^2}.$$

By Bayes' rule,

$$\mu_n(y) = \frac{q_A Q_{A,n}(y)}{q_A Q_{A,n}(y) + q_B Q_{B,n}(y)}.$$

Hence

$$\begin{aligned}
\mathbb{E}\left[\min\{\mu_n(Y_n), 1-\mu_n(Y_n)\}\right] &= \sum_y \min\{q_A Q_{A,n}(y), q_B Q_{B,n}(y)\} \\
&\geq \min\{q_A, q_B\} \sum_y \min\{Q_{A,n}(y), Q_{B,n}(y)\} \\
&\geq \frac{\min\{q_A, q_B\}}{2} e^{-48M^2} =: k > 0.
\end{aligned}$$

Let $W_n = \min\{\mu_n(Y_n), 1-\mu_n(Y_n)\}$. Since $0 \leq W_n \leq 1/2$ and $E[W_n] \geq k$, choosing $\varepsilon = k/2$ gives

$$\Pr(W_n \geq \varepsilon) \geq 2(k - \varepsilon) = k \geq \varepsilon$$

for all sufficiently large n. This proves the claim.

References

Austen-Smith, D. and Banks, J. S. (1996). Information aggregation, rationality, and the Condorcet jury theorem. *American Political Science Review*, **90** (1), 34–45.

Bhattacharya, S., Duffy, J. and Kim, S. (2017). Voting with endogenous information acquisition: Experimental evidence. *Games and Economic Behavior*, **102**, 316–338.

Bretagnolle, J. and Huber, C. (1979). Estimation des densités: risque minimax. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, **47** (2), 119–137.

Bulow, J. I., Geanakoplos, J. D. and Klemperer, P. D. (1985). Multimarket oligopoly: Strategic substitutes and complements. *Journal of Political Economy*, **93** (3), 488–511.

Downs, A. (1957). *An Economic Theory of Democracy*. New York: Harper & Row.

Dunning, T., Grossman, G., Humphreys, M., Hyde, S. D., McIntosh, C., Nellis, G., Adida, C. L., Arias, E., Bicalho, C., Boas, T. C. *et al.* (2019). Voter information campaigns and political accountability: Cumulative findings from a preregistered meta-analysis of coordinated trials. *Science Advances*, **5** (7).

Ekmekci, M., Heese, C. and Lauermann, S. (2025). *A generalized intermediate value theorem*. Working paper.

Elbittar, A., Gomberg, A., Martinelli, C. and Palfrey, T. R. (2020). Ignorance and bias in collective decisions. *Journal of Economic Behavior & Organization*, **174**, 332–359.

Feddersen, T. and Pesendorfer, W. (1997). Voting behavior and information aggregation in elections with private information. *Econometrica*, **65** (5), 1029–1058.

Feller, W. (1971). *An Introduction to Probability Theory and Its Applications*, vol. 2. New York: John Wiley & Sons, 2nd edn.

Gerardi, D. and Yariv, L. (2008). Information acquisition in committees. *Games and Economic Behavior*, **62** (2), 436–459.

Gershkov, A. and Szentes, B. (2009). Optimal voting schemes with costly information acquisition. *Journal of Economic Theory*, **144**, 36–68.

Gil de Zúñiga, H., Strauss, N. and Huber, B. (2020). The proliferation of the "news finds me" perception across societies. *International Journal of Communication*, **14**, 1605–1633.

—, Weeks, B. and Ardèvol-Abreu, A. (2017). Effects of the news-finds-me perception in communication: Social media use implications for news seeking and learning about politics. *Journal of Computer-Mediated Communication*, **22** (3), 105–123.

Grosser, J. and Seebauer, M. (2016). The curse of uninformed voting: An experimental study. *Games and Economic Behavior*, **97**, 205–226.

Heese, C. (2022). *Information frictions and opposing political interests*. Working paper.

Koriyama, Y. and Szentes, B. (2009). A resurrection of the condorcet jury theorem. *Theoretical Economics*, **4** (2), 227–252.

Martinelli, C. (2006). Would rational voters acquire costly information? *Journal of Economic Theory*, **129** (1), 225–251.

— (2007). Rational ignorance and voting behavior. *International Journal of Game Theory*, **35** (3), 315–335.

McCombs, M. and Poindexter, P. (1983). The duty to keep informed: News exposure and civic obligation. *Journal of Communication*, **33** (2), 88–96.

Mechtenberg, L. and Tyran, J.-R. (2019). Voter motivation and the quality of democratic choice. *Games and Economic Behavior*, **116**, 241–259.

MILGROM, P. and ROBERTS, J. (1990). Rationalizability, learning, and equilibrium in games with strategic complementarities. *Econometrica*, **58** (6), 1255–1277.

— and SHANNON, C. (1994). Monotone comparative statics. *Econometrica*, **62** (1), 157–180.

MIRANDA, C. (1940). *Un'osservazione su un teorema di Brouwer*. Consiglio Nazionale delle Ricerche.

MUKHOPADHAYA, K. (2003). Jury size and the free rider problem. *Journal of Law, Economics, and Organization*, **19** (1), 24–44.

MYERSON, R. B. (1998). Population uncertainty and Poisson games. *International Journal of Game Theory*, **27** (3), 375–392.

— (2009). Learning from Schelling's *Strategy of Conflict*. *Journal of Economic Literature*, **47** (4), 1109–25.

OLIVEROS, S. (2013). *Aggregation of endogenous information in large elections*. Working paper.

PALFREY, T. R. and ROSENTHAL, H. (1983). A strategic calculus of voting. *Public Choice*, **41** (1), 7–53.

— and — (1985). Voter participation and strategic uncertainty. *American Political Science Review*, **79** (1), 62–78.

PALMER, R. and TOFF, B. (2020). What does it take to sustain a news habit? the role of civic duty norms and a connection to a "news community" among news avoiders in the uk and spain. *International Journal of Communication*, **14**, 1634–1653.

PERSICO, N. (2004). Committee design with endogenous information. *The Review of Economic Studies*, **71** (1), 165–191.

PIKETTY, T. (1999). The information-aggregation approach to political institutions. *European Economic Review*, **43** (4–6), 791–800.

SCHELLING, T. C. (1980). *The Strategy of Conflict*. Harvard University Press.

TOFF, B. and KALOGEROPOULOS, A. (2020). All the news that's fit to ignore: How the information environment does and does not shape news avoidance. *Public Opinion Quarterly*, **84** (S1), 366–390.

TOPKIS, D. M. (1979). Equilibrium points in nonzero-sum n-person submodular games. *SIAM Journal on Control and Optimization*, **17** (6), 773–787.

— (1998). *Supermodularity and Complementarity*. Princeton, NJ: Princeton University Press.

TRIOSSI, M. (2013). Costly information acquisition. Is it better to toss a coin? *Games and Economic Behavior*, **82**, 169–191.

VIVES, X. (1990). Nash equilibrium with strategic complementarities. *Journal of Mathematical Economics*, **19** (3), 305–321.